

DNF-SR: Dual-Input and Negative-Aware Feature Fine-Tuning for Real-World Image Super-Resolution

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Abstract

Benefiting from the powerful generative priors of diffusion models, diffusion-based real-world image super-resolution (*Real-ISR*) methods have demonstrated impressive performance. To achieve efficient *Real-ISR*, several recent works have designed one-step diffusion-based models. However, unmediatedly feeding LR into a diffusion model creates a distributional gap with the model's original input. A straightforward approach to reduce the distribution gap is to introduce noise to the LR latents. However, directly adding noise inevitably corrupts the content of the LR images. In this study, we propose **DNF-SR**, a Dual-input and Negative-aware Feature fine-tuning method for *Real-ISR*. Specifically, we use a **dual-input** strategy that concatenates the original LR image with the noisy LR input and feeds them into a diffusion-based image editing model, ensuring both high-fidelity one-step super-resolution and improved perceptual and content consistency. Additionally, the noise present in the noisy LR input introduces randomness and diversity into the outputs. We exploit this property and propose a post-training optimization method, **Negative-aware Feature Fine-Tuning (NF²T)**, which guides the model toward producing higher-quality results. NF²T classifies multiple outputs into positive and negative subsets and then defines implicit policy improvement directions in both the image and feature spaces, thereby further enhancing the stability of the optimization. Extensive experiments show that DNF-SR outperforms other methods. Code is available at <https://github.com/SHH-Han/DNF-SR>.

1. Introduction

Image Super-Resolution (ISR) aims to reconstruct a clear high-resolution (HR) image from a degraded low-resolution

DNF-SR: 面向真实世界图像超分辨率的双输入与负向感知特征微调

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摘要

得益于扩散模型强大的生成先验，基于扩散的真实世界图像超分辨率 (*Real-ISR*) 方法已展现出令人瞩目的性能。为了实现高效的 *Real-ISR*，近期一些工作设计了单步基于扩散的模型。然而，直接将低分辨率图像输入扩散模型会产生与模型原始输入之间的分布差距。减小分布差距的一种直接方法是向低分辨率潜变量引入噪声。但是，直接添加噪声不可避免地会破坏低分辨率图像的内容。在本研究中，我们提出了 **DNF-SR**，一种面向真实图像超分辨率的双输入与负向感知特征微调方法。具体而言，我们采用一种双输入策略，将原始低分辨率图像与带噪低分辨率输入拼接后送入基于扩散的图像编辑模型，从而同时实现高保真的单步超分辨率以及更好的感知和内容一致性。此外，带噪低分辨率输入中存在的噪声为输出引入了随机性和多样性。我们利用这一特性，提出了一种后训练优化方法——负向感知特征微调 (NF²T)，引导模型生成更高质量的结果。NF²T 将多个输出划分为正负样本子集，然后在图像空间和特征空间中定义隐式策略改进方向，从而进一步增强优化的稳定性。大量实验表明，DNF-SR 优于其他方法。代码可在 <https://github.com/SHH-Han/DNF-SR> 获取。

1. 引言

图像超分辨率 (ISR) 旨在从退化的低分辨率图像中重建出清晰的高分辨率 (HR) 图像。

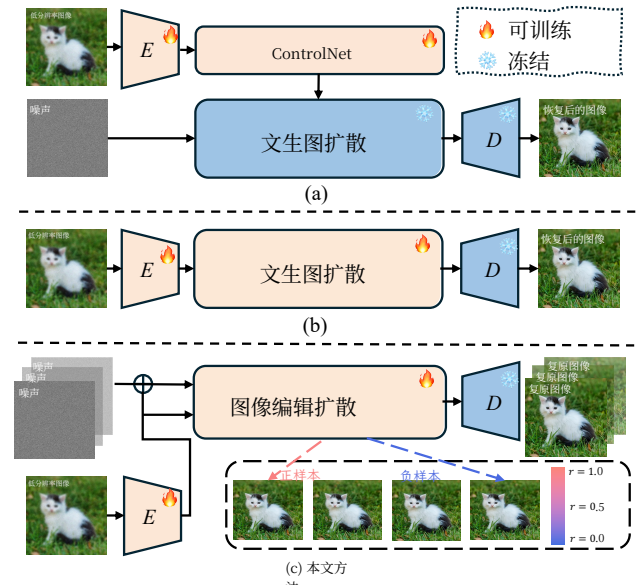


Figure 1. Different model architectures for one-step SR. (a) Utilizing ControlNet [62] to inject LR into a text-to-image (T2I) model, enabling one-step SR from noise directly to HR. (b) Directly using LR as input and fine-tuning a T2I model to generate HR. (c) We adopt a dual-input method, feeding noisy LR and original LR into the image editing model to generate HR, and utilize a Negative-aware Feature Fine-Tuning method for post-training.

(LR) image suffering from noise, blur, and other degradations. To better reconstruct low-quality real-world images that are affected by a wider range of degradations into more realistic images, many researchers have begun exploring the use of generative models [10, 35] for image super-resolution. Recently, as large-scale pretrained diffusion generative models [32, 34] have demonstrated impressive performance in image generation, an increasing number of studies [13, 24, 37, 50] have adopted techniques such as LoRA [18] or ControlNet [62] to transfer the powerful

(LR) 图像受到噪声、模糊及其他退化的影响。为了更好地将受更广泛退化影响的低质量真实世界图像重建为更真实感的图像，许多研究人员开始探索利用生成模型 [10, 35] 进行图像超分辨率。近年来，随着大规模预训练扩散生成模型 [32, 34] 在图像生成中展现出令人瞩目的性能，越来越多的研究 [13, 24, 37, 50] 采用 LoRA [18] 或ControlNet [62] 等技术，将扩散模型强大的

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high-quality image generation priors of diffusion models to Real-ISR, yielding more realistic and clearer restored images. Meanwhile, optimization under data constraints has also been explored [9].

Due to the multi-step denoising process inherent to diffusion models, they suffer from slow inference. To better leverage diffusion models for efficient super-resolution, recent works [12, 25, 39, 45, 49, 53, 60] have explored one-step diffusion-based SR models. As shown in Fig. 1, the current mainstream frameworks for one-step diffusion-based SR methods can be broadly categorized into the following two types. (a) Distilling multi-step super-resolution diffusion networks that use ControlNet (e.g., resshift [59], seesr [50]) into a one-step mapping from noise to high-resolution images, as in SinSR [45] and AddSR-1s [54]. These methods increase the parameter count by incorporating ControlNet. And the design that directly restores an HR image from noise in a single step limits the model’s performance. (b) Directly feeding the LR image into the diffusion model and fine-tuning the model with LoRA to reconstruct a clear image has been explored by works such as OSERDiff [49], TSDSR [12], and FluxSR [22]. These methods replace the diffusion model’s original noise input at the initial timesteps of the denoising process with the LR image latent, which introduces a substantial distributional gap in the model inputs and consequently degrades performance. Recent works such as OMGSR [53] and TADSR [63], which still follow the (b) architecture, explore using intermediate timesteps to perform one-step diffusion super resolution. However, both the HR latent and the noise contain more high-frequency information than the LR latent. Therefore, at any intermediate timestep, the LR latent differs from the diffusion model’s original input in terms of high-frequency information. Adding noise to LR is the most straightforward method to reduce the distribution gap, but it may destroy the content of LR.

To address the issues above, we propose **DNF-SR**, a **D**ual-input and **N**egative-aware **F**eature **F**ine-Tuning method for Real-ISR, with its framework shown in Fig. 1(c). Specifically, we design a **dual-input** strategy that inputs noisy LR to eliminate the gap with the original input of the Diffusion model, while additionally incorporating the original LR as a condition to ensure fidelity. And using Flux-Kontext [4] as a pretrained diffusion model, we patchify both the noisy LR and the original LR, then concatenate them along the token dimension and feed these tokens into the DiT [31] blocks for image super-resolution. With this dual-input design, our method reduces the input distribution gap of the diffusion model, more effectively leverages the generative prior of the diffusion model, and preserves the fidelity of the LR image. Rather than employing the text-to-image generation model Flux, using an editing model can better perceive the original LR content as

a condition, thereby improving the quality of outputs.

Due to the introduction of noise, our one-step SR model can generate diverse restored images. To further improve the quality of generated images, we propose a novel super-resolution post-training method named **Negative-aware Feature Fine-Tuning**, named **NF²T**. NF²T draws on the approach of DiffusionNFT [64] by partitioning the multiple generated images into positive and negative subsets according to a reward model and then defining an implicit policy improvement direction to enhance model performance. However, we observed that applying DiffusionNFT to the super-resolution task introduces pronounced grid artifacts in the reconstructed images due to the implicit optimization direction in the latent space. Therefore, we adopt NF²T, transferring the optimization from the latent space to the image and the feature spaces for improved performance. Compared with other reinforcement learning methods [27, 41], NF²T omits probabilistic modeling in sampling and optimization, improving generative model performance by directly optimizing the predicted velocity in flow matching. Meanwhile, NF²T can be optimized exploiting multiple sampled images, unlike DiffusionDPO, which only uses image pairs. During post-training, we evaluate the multiple generated images using several IQA methods [7, 11, 21, 42, 57], including full reference (FR) and no reference (NR) metrics, and then aggregate the normalized scores into a single reward for optimization. The use of diverse metrics provides clearer optimization signals for the model and thereby enables recovery of higher quality images.

- The main contributions of our work are as follows:
- We design a dual-input strategy that feeds both noisy LR and original LR into an image-editing model, which can effectively capture conditional LR image information. This approach better leverages the model’s generative prior while ensuring higher fidelity, thereby further enhancing the performance of SR model.
- We propose a novel SR optimization method named **Negative-aware Feature Fine-Tuning**. It categorizes multiple sampled restored images into positive and negative optimization directions within the image and feature spaces, which further improves the quality of outputs.

2. Related Work

Real-World Image Super-Resolution. Recently, large-scale pretrained diffusion generative models [32, 34] have achieved strong results in Real-ISR tasks [13, 24, 36, 37, 44, 50, 51, 56], yielding more realistic and clearer restored images. However, the multi-step denoising process results in high computational and time costs. Consequently, one-step Real-ISR models have become a primary research focus for faster inference. OSERDiff [49] introduces the VSD loss to distill the pre-trained SD model. Building upon this, TSD-

高质量图像生成先验迁移到真实图像超分辨率中，从而产生更真实、更清晰的复原图像。同时，数据约束下的优化也已被探索 [9]。

由于扩散模型固有的多步去噪过程，它们面临推理速度慢的问题。为了更好地利用扩散模型实现高效超分辨率，近期研究 [12, 25, 39, 45, 49, 53, 60] 探索了单步基于扩散的超分辨率模型。如图1所示，当前单步基于扩散的超分辨率方法的主流框架大致可分为以下两类。（a）将使用ControlNet的多步超分辨率扩散网络（如resshift [59],seesr [50]) 蒸馏为从噪声到高分辨率图像的单步映射，如SinSR [45] 和 AddSR-1s [54]。这些方法通过引入ControlNet增加了参数量。而且，从噪声中单步直接恢复高分辨率图像的设计限制了模型的性能。（b）直接将低分辨率图像输入扩散模型，并使用LoRA对模型进行微调以重建清晰图像，已有OSERDiff [49], TSDSR [12], 和 FluxSR [22]等工作对此进行了探索。这些方法在去噪过程的初始时间步用低分辨率图像潜变量替换了扩散模型原本的噪声输入，这在模型输入中引入了显著的分布差距，从而导致性能下降。近期工作如OMGSR [53]和TADSR[63], 仍沿用（b）的架构，探索使用中间时间步来执行单步扩散超分辨率。然而，高分辨率潜变量和噪声都比低分辨率潜变量包含更多高频信息。因此，在任何中间时间步，低分辨率潜变量在高频信息方面都与扩散模型的原始输入存在差异。向低分辨率图像添加噪声是减小分布差距最直接的方法，但这可能会破坏低分辨率图像的内容。

为了解决上述问题，我们提出了**DNF-SR**，一种用于真实图像超分辨率的**双输入**和**负感知特征**微调方法，其框架如图1(c)所示。具体来说，我们设计了一种**双输入**策略，输入含噪低分辨率图像以消除与扩散模型原始输入之间的差距，同时额外引入原始低分辨率图像作为条件以确保保真度。并且使用Flux-Kontext [4]作为预训练扩散模型，我们将含噪低分辨率图像和原始低分辨率图像都进行分块处理，然后沿token维度将它们拼接起来，并将这些token输入到DiT [31] 模块中进行图像超分辨率。通过这种双输入设计，我们的方法减小了扩散模型的输入分布差距，更有效地利用了扩散模型的生成先验，并保持了低分辨率图像的保真度。与采用文本到图像生成模型Flux相比，使用编辑模型能够更好地感知原始低分辨率图像内容作为

条件，从而提高输出质量。

由于引入了噪声，我们的单步超分辨率模型能够生成多样的恢复图像。为进一步提升生成图像的质量，我们提出了一种新的超分辨率后训练方法，称为**Negative-awareFeature Fine-Tuning**，即**NF²T**。NF²T借鉴了DiffusionNFT [64] 的思路，根据奖励模型将多个生成图像划分为正样本和负样本量子集，然后定义隐式策略改进方向以提升模型性能。然而，我们观察到将DiffusionNFT应用于超分辨率任务时，由于潜空间中的隐式优化方向，会在重建图像中引入明显的网格伪影。因此，我们采用NF²T，将优化从潜空间转移到图像空间和特征空间，以获得更好的性能。与其他强化学习方法相比， [27, 41], NF²T省略了采样和优化中的概率建模，通过直接优化流匹配中的预测速度来提升生成模型的性能。同时，与仅使用图像对的DiffusionDPO不同，NF²T可以利用多个采样图像进行优化。在后训练过程中，我们使用多种图像质量评估方法[7, 11, 21, 42, 57], （包括全参考（FR）和无参考（NR）评价指标）评估多个生成图像，然后将归一化后的分数聚合为单一奖励用于优化。使用多样化的评价指标为模型提供了更清晰的优化信号，从而能够恢复出更高质量的图像。

- 我们工作的主要贡献如下：
- 我们设计了一种双输入策略，将带噪声的低分辨率图像和原始低分辨率图像同时输入图像编辑模型，从而有效捕获条件低分辨率图像信息。该方法能更好地利用模型的生成先验，同时确保更高的保真度，从而进一步提升超分辨率模型的性能。
- 我们提出了一种新颖的超分辨率优化方法，命名为负向感知特征微调。它在图像与特征空间中多个采样的恢复图像划分为正、负优化方向，进一步提高输出质量。

2. 相关工作

真实世界图像超分辨率。近年来，大规模预训练扩散生成模型 [32, 34] 在真实图像超分辨率任务中取得了显著成果 [13, 24, 36, 37, 44,50, 51, 56], , 能够生成更逼真、更清晰的恢复图像。然而，多步去噪过程带来了高昂的计算和时间成本。因此，单步真实图像超分辨率模型已成为加快推理速度的主要研究方向。OSERDiff [49] 引入了变分分数蒸馏损失来蒸馏预训练的稳态扩散模型。在此基础上，TSD-

SR [12] designs a Target Score Distillation for SR. SinSR [45] achieves one-step ISR from noise to HR by distilling a multi-step diffusion model into a student network through deterministic mapping. OMGSR [53] injects the LR image latent distribution at a pre-computed mid-timestep, achieving the state-of-the-art performance. Nevertheless, there is a gap between the LR latent and the diffusion model’s original input in terms of high-frequency information. To eliminate this gap, we design dual-input methods, inputting noisy LR and incorporating original LR as a condition. Furthermore, an image-editing model is employed to leverage this dual-path input better, thereby producing higher-quality outputs.

Preference Alignment for Diffusion Models. Since the proposal of GRPO [17], Reinforcement Learning from Human Feedback (RLHF) [3, 30] has emerged as one of the most prominent and widely studied research topics. Diffusion models and rectified flows can also benefit significantly from alignment with human feedback for their generation diversity. Many methods are based on likelihood estimation, including: (1) Policy gradient methods from PPO-style [5, 14] decompose trajectory likelihoods step-wise without forward consistency, while recent GRPO extensions [27, 55] convert ODE to SDE samplers, proving effective and scalable for diffusion RL. However, this method is hard to apply to one-step Real-ISR diffusion since it requires a multi-step denoising process to guarantee diversity. (2) Direct Policy Optimization (DPO)-style [33] methods. Diffusion-DPO [41] adapts DPO to diffusion, and it is inherently confined to paired human preference data, whose annotation process is labor-intensive and expensive. Moreover, its pairwise training paradigm hinders the exploitation of accurate ranking information among multiple generated results, thereby limiting the speed of optimization. Recently, DiffusionNFT [64] eliminates the reliance on likelihood estimation and SDE-based reverse process by formulating policy improvement as a contrast between positive and negative generations. It fully utilizes information from multiple sampled results to determine a better optimization direction. However, directly applying DiffusionNFT to SR tasks introduces a pronounced grid artifact. To address this issue, we propose a novel Negative-aware feature fine-tune (NF²T) method.

3. Methodology

3.1. Preliminaries

Flow-matching Model in Flux. As a core generative model used by Flux [6], flow matching (FM) [26] aims to learn a continuous normalizing flow that maps a simple distribution (e.g., standard Gaussian distribution $\epsilon \in \mathcal{N}(0, \mathbf{I})$) to the target data distribution $x_0 \sim \pi_0 = p_{data}$. The specific training

loss function is as follows:

$$\mathcal{L} = \mathbb{E}_{x_0 \sim \pi_0, \epsilon \sim \mathcal{N}(0, \mathbf{I}), t} \|v_\theta(x_t, t) - v\|^2, \quad (1)$$

where v denotes the velocity at any time $t \in [0, 1]$ predicted by the model. Flux adopts the same method from Rectified Flow [28], which assumes a simple linear path $x_t = (1-t)x_0 + t\epsilon$, thereby deriving the optimization target for velocity as $v = \epsilon - x_0$.

DiffusionNFT for Preference Alignment. DiffusionNFT adopts a novel Preference Alignment Paradigm for diffusion models to achieve efficient alignment with human or task-specific preferences. The specific approach of DiffusionNFT involves computing the reward r for multiple sampled data from the diffusion model, normalizing these rewards to the range $[0, 1]$, then leveraging positive and negative sample signals to implicitly model preference directions. The specific training objective is as follows:

$$\mathcal{L} = \mathbb{E}_{x_0 \sim \pi_{old}(x_0|c), \epsilon \sim \mathcal{N}(0, \mathbf{I}), c, t} \left[r \|v_\theta^+(x_t, c, t) - v\|^2 + (1-r) \|v_\theta^-(x_t, c, t) - v\|^2 \right], \quad (2)$$

where implicit positive policy:

$$v_\theta^+(x_t, c, t) := (1 - \beta)v^{\text{old}}(x_t, c, t) + \beta v_\theta(x_t, c, t),$$

and implicit negative policy:

$$v_\theta^-(x_t, c, t) := (1 + \beta)v^{\text{old}}(x_t, c, t) - \beta v_\theta(x_t, c, t).$$

In Eq. (2), π_{old} denotes the pretrained diffusion policy, v_{old} and v_θ represent the velocity predictor of the pretrained diffusion and the optimized diffusion model, respectively, c denotes conditional control, and β is a hyperparameter representing the guidance strength.

3.2. Overview of DNF-SR

As shown in Fig. 2, DNF-SR mainly consists of two parts. One part is the **model structure of DNF-SR**, where we use Flux-kontext as a pre-trained model and design a single-step super-resolution model with a dual-input at a mid-timestep t_{mid} . The other part is the **post-training of DNF-SR**, where we adopt a Negatively-aware Feature Fine-Tuning (NF²T) method, achieving preference optimization by dividing sampled data into positive and negative optimization direction in both image and feature spaces.

3.3. Model Structure of DNF-SR

The model structure of DNF-SR is shown in Fig. 2(a). The inputs to DNF-SR are the LR image x_{LR} , its caption c , and randomly initialized noise ϵ . First, x_{LR} is fed into a fine-tuned Encoder E_θ to obtain z_{LR} . Then z_{LR} and ϵ are weighted according to a fixed mid-timestep t_{mid} to get $z_{mix} = (1 - t_{mid})z_{LR} + t_{mid}\epsilon$. Next, DNF-SR feeds the

SR [12] 为超分辨率设计了目标得分蒸馏。SinSR[45] 通过确定性映射将多步扩散模型蒸馏为学生网络，实现了从噪声到高分辨率的一步图像超分辨率。

OMGSR [53] 在预先计算的中间时间步注入低分辨率图像潜变量分布，达到了最先进的性能。然而，低分辨率潜变量与扩散模型原始输入在高频信息方面仍存在差距。为消除这一差距，我们设计了双输入方法，既输入含噪低分辨率图像，又将原始低分辨率图像作为条件引入。此外，我们采用图像编辑模型来更好地利用这种双路径输入，从而生成更高质量的输出。

扩散模型的偏好对齐。自GRPO被提出以来 [17], , 基于人类反馈的强化学习（RLHF） [3, 30] 已成为最突出且被广泛研究的热点之一。扩散模型和整流流也能从与人类反馈的对齐中显著受益，从而提升其生成的多样性。许多方法基于似然估计，包括：（1）源自PPO风格的策略梯度方法 [5, 14]在不具备前向一致性的情况下逐步分解轨迹似然，而近期的GRPO扩展 [27, 55] 将常微分方程转换为随机微分方程采样器，已被证明对扩散强化学习有效且可扩展。然而，该方法难以应用于单步真实图像超分辨率扩散，因为它需要多步去噪过程来保证多样性。（2）直接策略优化（DPO）风格 [33] 方法。Diffusion-DPO [41]将DPO适配到扩散模型中，但其本质上受限于成对人类偏好数据，而此类数据的标注过程劳动密集且成本高昂。此外，其成对训练范式阻碍了对多个生成结果之间精确排序信息的利用，从而限制了优化速度。近期，DiffusionNFT [64] 通过将策略改进表述为正样本与负样本生成之间的对比，消除了对似然估计和基于随机微分方程的逆向过程的依赖。它充分利用多个采样结果中的信息来确定更优的优化方向。然而，将DiffusionNFT直接应用于超分辨率任务会引入明显的网格伪影。为解决这一问题，我们提出了一种新颖的负样本感知特征微调（NF²T）方法。

3. 方法论

3.1. 预备知识

Flux中的流匹配模型。作为Flux [6], 使用的核心生成模型，流匹配（FM） [26] 旨在学习一个连续归一化流，将简单分布（例如标准高斯分布 $\epsilon \in \mathcal{N}(0, \mathbf{I})$ ）映射到目标数据分布 $x_0 \sim \pi_0 = p_{data}$ 。具体训练

损失函数如下：

$$\mathcal{L} = \mathbb{E}_{x_0 \sim \pi_0, \epsilon \sim \mathcal{N}(0, \mathbf{I}), t} \|v_\theta(x_t, t) - v\|^2, \quad (1)$$

其中 v 表示模型在任意时刻 $t \in [0, 1]$ 预测的速度。Flux 采用了与矫正流 [28], 相同的方法，该方法假设一条简单的线性路径 $x_t = (1-t)x_0 + t\epsilon$ ，从而推导出速度的优化目标为 $v = \epsilon - x_0$ 。

用于偏好对齐的 DiffusionNFT。DiffusionNFT 为扩散模型采用了一种新颖的偏好对齐范式，以实现与人类或任务特定偏好的高效对齐。DiffusionNFT 的具体方法包括计算扩散模型多个采样数据的奖励 r ，将这些奖励归一化到 $[0, 1]$ 范围，然后利用正样本和负样本信号来隐式建模偏好方向。具体的训练目标如下：

$$\mathcal{L} = \mathbb{E}_{x_0 \sim \pi_{old}(x_0|c), \epsilon \sim \mathcal{N}(0, \mathbf{I}), c, t} \left[r \|v_\theta^+(x_t, c, t) - v\|^2 + (1-r) \|v_\theta^-(x_t, c, t) - v\|^2 \right], \quad (2)$$

其中隐式正策略：

$$v_\theta^+(x_t, c, t) := (1 - \beta)v^{\text{old}}(x_t, c, t) + \beta v_\theta(x_t, c, t),$$

隐式负策略：

$$v_\theta^-(x_t, c, t) := (1 + \beta)v^{\text{old}}(x_t, c, t) - \beta v_\theta(x_t, c, t).$$

在公式 (2) 中， π_{old} 表示预训练扩散策略， v_{old} 和 v_θ 分别表示预训练扩散模型与优化扩散模型的速度预测器， c 表示条件控制， β 是表示引导强度的超参数。

3.2. DNF-SR概览

如图2所示，DNF-SR主要由两部分组成。一部分是**DNF-SR的模型结构**，其中我们使用Flux-Kontext作为预训练模型，并设计了一个在中间时间步采用双输入的单步超分辨率模型 t_{mid} 。另一部分是**DNF-SR的后训练**，其中我们采用负感知特征微调（NF²T）方法，通过在图像与特征空间中将采样数据分为正优化方向和负样本优化方向来实现偏好优化。

3.3. DNF-SR的模型结构

DNF-SR 的模型结构如图2(a) 所示。DNF-SR 的输入是低分辨率图像 x_{LR} 、其文本描述 c 以及随机初始化的噪声 ϵ 。首先， x_{LR} 被输入微调后的编码器 E_θ 以得到 z_{LR} 。然后， z_{LR} 和 ϵ 根据固定中间时间步 t_{mid} 进行加权，得到 $z_{mix} = (1 - t_{mid}) z_{LR} + t_{mid}\epsilon$ 。接下来，DNF-SR 将

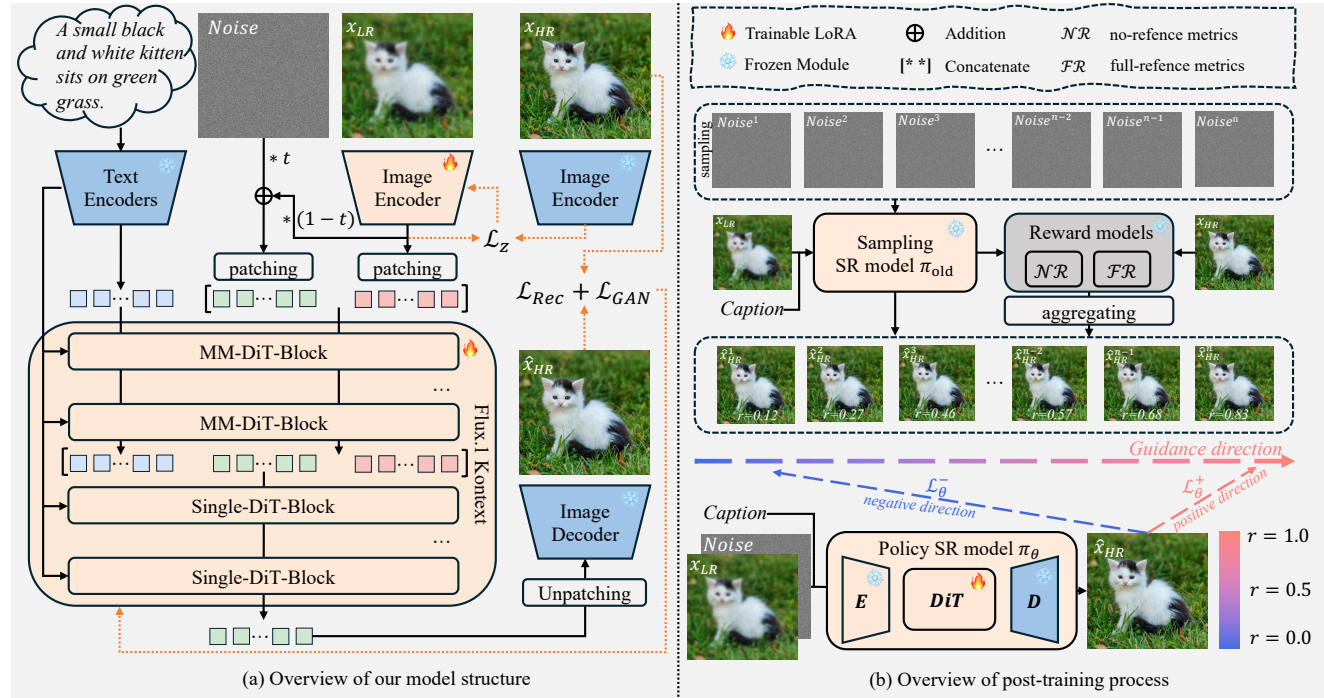


Figure 2. Overview of the DNF-SR framework. (a) The model structure adopts a dual-input design: noisy LR latents (z_{mix}) and original LR latents (z_{LR}) are concatenated with text tokens (encoded from image captions) and fed into fine-tuned DiT blocks of the Flux-Kontext pretrained model. We employed multiple loss functions to ensure the performance of the model. (b) The post-training process of Negative-aware Feature Fine-Tuning: multiple restored results are generated with different input noises, evaluated by combined full-reference (FR) and no-reference (NR) IQA metrics to obtain reward scores, and divided into positive and negative subsets. By constructing positive $\mathcal{L}_\theta^+$ and negative $\mathcal{L}_\theta^-$ optimization directions within both the image and the feature spaces, the quality of the restored images is improved.

concatenated z_{mix} and z_{LR} along with the text condition z_{text} (encoded using c) into fine-tuned DiT blocks, as employed in Flux-Kontext. Specifically, all tokens from z_{mix} , z_{LR} , and z_{text} are concatenated after passing through multiple MM-DiT blocks. This combined sequence then goes through several Single-DiT blocks. Subsequently, the output at the position corresponding to z_{mix} is extracted as the model's predicted velocity v_θ , which is then used to obtain $\hat{z}_{HR} = z_{mix} - t_{mid}v_\theta$. Finally, $\hat{z}_{HR}$ is passed through a fixed-parameter decoder D_φ to obtain $\hat{x}_{HR}$.

In comparison to other methods, our primary improvements involve employing a **dual input for SR** and performing a **one-step denoising at mid-timestep**.

Dual-input for SR. Most current single-step SR methods directly feed the LR image x_{LR} into a Diffusion model to obtain the HR image $\hat{x}_{HR}$. Compared to the original noise input of Diffusion models, there's a significant distribution gap between the LR data and noise. Directly replacing the input makes it difficult for Diffusion's pre-trained parameters to be effectively applied to SR tasks. Although recent works [53, 63] have explored replacing the input at mid-timestep to mitigate this gap between LR and the Diffusion model's original input, this introduces a new question: At

which timestep t does the LR latent z_{LR} have a smaller distribution difference with the original Diffusion model's latent input z_t ? In fact, from a frequency perspective, z_{LR} is most consistent with the latent representation of a natural image z_0 at $t = 0$, rather than an arbitrary mid-timestep's z_t . Relative to x_0 , $x_t = (1 - t)x_0 + t\epsilon$ exhibits richer high-frequency components owing to noise injection. For super-resolution, low-resolution (LR) images contain fewer high-frequency components than high-resolution (HR) images x_0 . Thus, x_0 is distributionally more similar to LR images in the diffusion framework. However, in SR tasks, if $t = 0$, the model would be unable to optimize effectively due to $\hat{z}_{HR} = z_{LR} - t v_\theta$. To address this issue and better eliminate the distribution gap between z_{LR} and z_t , one straightforward method is adding noise to z_{LR} to obtain z_{mix} as the input to the model. However, it can corrupt the content of the LR images. Therefore, to avoid content degradation caused by adding noise to z_{LR} , we design a dual-input method, which feeds the original z_{LR} as a conditional control along with z_{mix} into the Diffusion model. Meanwhile, unlike other single-step super-resolution approaches, which use text-to-image models as the pretrained backbone, we employ a pretrained image-editing model. This enables

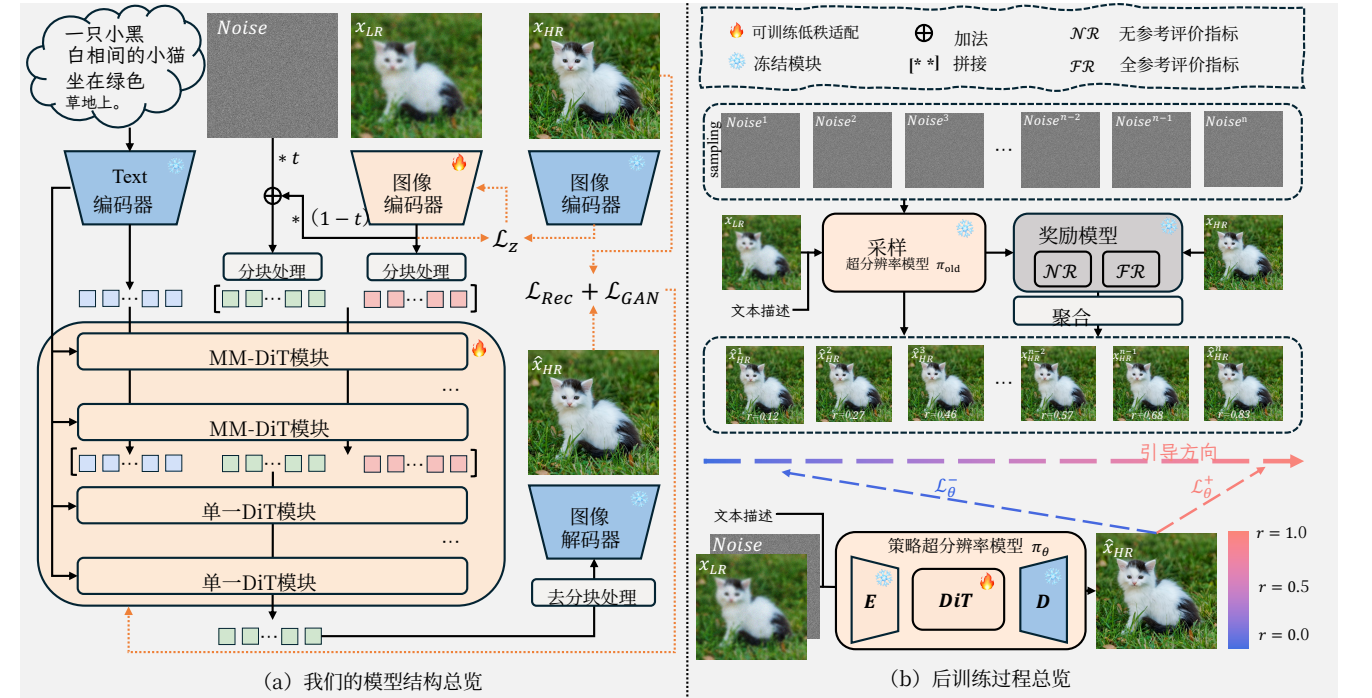


Figure 2. DNF-SR 框架总览。(a) 模型结构采用双输入设计：带噪低分辨率潜变量 (z_{mix}) 和原始低分辨率潜变量 (z_{LR}) 与文本标记 (由图像描述编码而来) 拼接后，输入 Flux-Kontext 预训练模型中经过微调的 DiT 块。我们采用了多种损失函数来确保模型性能。(b) 负样本感知特征微调的后训练过程：使用不同输入噪声生成多个恢复结果，通过全参考 (FR) 和无参考 (NR) 图像质量评价指标组合进行评估以获得奖励分数，并划分为正负样本子集。通过在图像空间和特征空间中构建正样本 $\mathcal{L}_\theta^+$ 与负样本 $\mathcal{L}_\theta^-$ 优化方向，恢复图像的质量得到提升。

拼接后的 z_{mix} 和 z_{LR} 与文本条件 z_{text} (使用 c 编码) 一同输入微调后的 DiT 模块，正如 Flux-Kontext 中所采用的方式。具体而言，来自 z_{mix} 、 z_{LR} 和 z_{text} 的所有 token 在经过多个 MM-DiT 块后被拼接在一起。该组合序列随后通过若干个 Single-DiT 块。然后，在 z_{mix} 对应位置上的输出被提取为模型的预测速度 v_θ ，并用其得到 $\hat{z}_{HR} = z_{mix} - t_{mid}v_\theta$ 。最后， $\hat{z}_{HR}$ 通过固定参数解码器 D_φ 以获得 $\hat{x}_{HR}$ 。

与其他方法相比，我们的主要改进在于采用**用于超分辨率的双输入**，并在**中间时间步执行一步去噪**。

用于超分辨率的双输入。当前大多数单步超分辨率方法直接将低分辨率图像 x_{LR} 输入扩散模型以获得高分辨率图像 $\hat{x}_{HR}$ 。与扩散模型原始的噪声输入相比，低分辨率数据与噪声之间存在显著分布差距。直接替换输入使扩散模型的预训练参数难以有效应用于超分辨率任务。尽管近期的研究 [53, 63] 已探索在中间时间步替换输入以缓解低分辨率与扩散模型原始输入之间的差距，但这引入了一个新问题：在

哪个时间步 t 下，低分辨率潜变量 z_{LR} 与原始扩散模型的潜变量输入之间的分布差异较小 z_t ? 事实上，从频率视角来看， z_{LR} 与自然图像的潜变量表示在 $t = 0$ 处最为一致，而不是任意中间时间步 z_{t_0} 。相对于 x_0 ， $x_t = (1 - t)x_0 + t\epsilon$ 由于噪声注入而表现出更丰富的高频分量。对于超分辨率来说，低分辨率图像比高分辨率图像包含更少的高频分量 x_0 。因此，在扩散框架中， x_0 在分布上更接近低分辨率图像。然而，在超分辨率任务中，如果 $t = 0$ ，由于 $\hat{z}_{HR} = z_{LR} - t v_\theta$ ，模型将无法有效优化。为了解决这个问题并更好地消除 z_{LR} 与 z_t 之间的分布差距，一种直接的方法是向 z_{LR} 添加噪声以获得 z_{mix} 作为模型的输入。然而，这会破坏低分辨率图像的内容。因此，为了避免向 z_{LR} 添加噪声所导致的内容退化，我们设计了一种双输入方法，将原始 z_{LR} 作为条件控制与 z_{mix} 一起馈入扩散模型。同时，与其他使用文本到图像模型作为预训练骨干网络的单步超分辨率方法不同，我们采用预训练的图像编辑模型。这使得

more effective utilization of the original LR image as a conditioning signal, thereby further enhancing super-resolution performance. Through this approach, we better leverage the model’s pre-trained parameters while maintaining image fidelity, generating high-quality $\hat{x}_{HR}$.

One-step denoising at mid-timestep. Diffusion models focus on generating different frequency components at various timesteps. Specifically, as t decreases, the model’s attention shifts from generating low-frequency information (i.e., the overall structure of the image) towards prioritizing the generation of high-frequency information (i.e., fine-grained details and textures). To ensure both fidelity and visual quality, we choose the latent z_{mix} corresponding to an intermediate timestep t_{mid} as the model input, rather than pure noise ϵ .

Training Objective. As shown in Fig. 2, the loss functions during the fine-tuning stage of DNF-SR consist of three components: $\mathcal{L}_z$, $\mathcal{L}_{Rec}$, and $\mathcal{L}_{GAN}$. Specifically, $\mathcal{L}_z$ is designed to align the LR latent z_{LR} , after passing through the fine-tuned Encoder E_θ , with the origin HR latent z_{HR} . This is formulated as:

$$\mathcal{L}_z = \mathcal{L}_{MSE}(z_{LR}, z_{HR}). \quad (3)$$

$\mathcal{L}_{Rec}$ represents the reconstruction loss between the restored $\hat{x}_{HR}$ and the ground-truth HR image x_{HR} , calculated as:

$$\mathcal{L}_{Rec}(\hat{x}_{HR}, x_{HR}) = \mathcal{L}_{MSE}(\hat{x}_{HR}, x_{HR}) + \mathcal{L}_{DISTS}(\hat{x}_{HR}, x_{HR}). \quad (4)$$

$\mathcal{L}_{GAN}$ is the generative adversarial [15], where we employ DINOv3 as the discriminator. Ultimately, the total loss is:

$$\mathcal{L}_{sft} = \lambda_1 \mathcal{L}_z + \lambda_2 \mathcal{L}_{MSE} + \lambda_3 \mathcal{L}_{DISTS} + \lambda_4 \mathcal{L}_{GAN}. \quad (5)$$

3.4. Post-training of DNF-SR

To further enhance the quality of the reconstructed images, we adopt DiffusionNFT, a preference optimization method based on Negative-aware Fine-Tuning (NFT), to improve model performance.

When applying DiffusionNFT, according to Eq. (2), we set the guidance strength $\beta = 1$ and use a fixed mid-timestep t_{mid} , resulting in the NFT optimization objective for single-step super-resolution as:

$$\mathcal{L}_{NFT} = \mathbb{E}_{\pi_{old}(\hat{z}_{HR}|z_{LR}), z_{LR}} \left[r ||v_\theta^+(z_{mix}, z_{LR}) - v||^2 + (1-r) ||v_\theta^-(z_{mix}, z_{LR}) - v||^2 \right], \quad (6)$$

where $z_{mix} = t_{mid}\epsilon + (1 - t_{mid})z_{LR}$, $v = (z_{mix} - \hat{z}_{HR})/t_{mid}$, $v_\theta^+ = v_\theta$, $v_\theta^- = 2v_{old} - v_\theta$ and r represents

the reward of $\hat{x}_{HR}$. Simultaneously, we define $\mathcal{L}_\theta^+$ to represent the positive optimization objective and $\mathcal{L}_\theta^-$ to represent the negative optimization objective:

$$\mathcal{L}_\theta^+ = r ||v_\theta^+ - v||^2, \mathcal{L}_\theta^- = (1-r) ||v_\theta^- - v||^2. \quad (7)$$

Feature Space Optimization. We observe that directly using $\mathcal{L}_\theta^+$ and $\mathcal{L}_\theta^-$ as optimization objective led to noticeable grid artifacts in the output images of the optimized model. Therefore, we transform the optimization in the velocity space to the image space using a function $f(v) = D_\varphi(z_{mix} - t_{mid}v)$. $v_\theta^+, v_\theta^-, v$ are fed into $f(v)$ to obtain $\hat{x}_\theta^+, \hat{x}_\theta^-, \hat{x}_{HR}$. Then, by utilizing the $\mathcal{L}_{rec}$ from the SFT training phase, we derive new positive optimization objectives $\mathcal{L}_\theta^{' +}$ and negative optimization objectives $\mathcal{L}_\theta^{' -}$ in image and feature space:

$$\mathcal{L}_\theta^{' +} = r \mathcal{L}_{Rec}(\hat{x}_\theta^+, \hat{x}_{HR}), \mathcal{L}_\theta^{' -} = (1-r) \mathcal{L}_{Rec}(\hat{x}_\theta^-, \hat{x}_{HR}). \quad (8)$$

This results in the final training objective for DNF-SR as follows:

$$\mathcal{L}_{NF^2T} = \mathbb{E}_{\pi_{old}(\hat{z}_{HR}|z_{LR}), z_{LR}} \left[r \mathcal{L}_{Rec}(\hat{x}_\theta^+, \hat{x}_{HR}) + (1-r) \mathcal{L}_{Rec}(\hat{x}_\theta^-, \hat{x}_{HR}) \right]. \quad (9)$$

Unlike reinforcement learning methods that require explicit probability modeling, NF²T directly enhances model performance through implicit optimization directions. This characteristic makes it particularly well-suited for one-step SR tasks. Furthermore, in contrast to DiffusionDPO, NF²T can leverage multiple sampled images to determine the optimization direction, whereas DiffusionDPO is limited to pairwise preference data, which restricts data utilization efficiency during training.

Reward Calculation. Regarding the calculation of rewards, for the K samples generated during each optimization step, we employ multiple evaluation metrics as rewards to ensure the accuracy of the optimization direction. The evaluation metrics used are broadly categorized into full-reference and no-reference metrics. Full-reference (FR) metrics include LPIPS [7] and DISTS [11], while no-reference (NR) metrics include CLIPQA [42], MUSIQ [21] and MANIQA [57]. After computing r_i^{raw} for the i -th metric, we standardize the rewards within the batch to obtain r_i^{std} . Assuming r_i^{std} follows a standard Gaussian distribution, we obtain the normalized reward r_i by calculating $r_i = \Phi(X < r_i^{std})$, where $\Phi(\cdot)$ is the standard Gaussian cumulative distribution function. Finally, the r_i values from different metrics are averaged to yield the ultimate reward.

4. Experiments

4.1. Experimental Settings.

Training Datasets. Following DiT4SR [13], we adopt DIV2K [1], DIV8K [16], Flickr2K [40], NKUSR8K [13],

能够更有效地利用原始低分辨率图像作为条件信号, 从而进一步提升超分辨率性能。通过这种方法, 我们能够在保持图像保真度的同时更好地利用模型的预训练参数, 生成高质量的 $\hat{x}_{HR}$ 。

中间时间步的一步去噪。 扩散模型在不同时间步上侧重生不同的频率分量。具体而言, 随着 t 减小, 模型的注意力会从生成低频信息 (即图像的整体结构) 转向优先生成高频信息 (即细粒度细节和纹理)。为了同时保证保真度和视觉质量, 我们选择与中间时间步 t_{mid} 对应的潜变量 z_{mix} 作为模型输入, 而不是纯噪声 ϵ 。

训练目标。 如图2所示, DNF-SR微调阶段的损失函数由三个部分组成: $\mathcal{L}_z$ 、 $\mathcal{L}_{Rec}$ 和 $\mathcal{L}_{GAN}$ 。具体而言, $\mathcal{L}_z$ 旨在使低分辨率潜变量 z_{LR} 在通过微调后的编码器 E_θ 之后, 与原始高分辨率潜变量 z_{HR} 对齐。其公式如下:

$$\mathcal{L}_z = \mathcal{L}_{MSE}(z_{LR}, z_{HR}). \quad (3)$$

$\mathcal{L}_{Rec}$ 表示恢复得到的 $\hat{x}_{HR}$ 与真实高分辨率图像 x_{HR} 之间的重建损失, 计算公式为:

$$\mathcal{L}_{Rec}(\hat{x}_{HR}, x_{HR}) = \mathcal{L}_{MSE}(\hat{x}_{HR}, x_{HR}) + \mathcal{L}_{DISTS}(\hat{x}_{HR}, x_{HR}). \quad (4)$$

$\mathcal{L}_{GAN}$ 是生成对抗 [15], 其中我们采用DINOv3作为判别器。最终, 总损失为:

$$\mathcal{L}_{sft} = \lambda_1 \mathcal{L}_z + \lambda_2 \mathcal{L}_{MSE} + \lambda_3 \mathcal{L}_{DISTS} + \lambda_4 \mathcal{L}_{GAN}. \quad (5)$$

3.4. DNF-SR的后训练

为了进一步提升重建图像的质量, 我们采用 DiffusionNFT——一种基于负向感知微调 (NFT) 的偏好优化方法——来改善模型性能。

在应用DiffusionNFT时, 根据式 (2), 我们设置引导强度 $\beta = 1$ 并使用固定的中间时间步 t_{mid} , 从而得到单步超分辨率的NFT优化目标为:

$$\mathcal{L}_{NFT} = \mathbb{E}_{\pi_{old}(\hat{z}_{HR}|z_{LR}), z_{LR}} \left[r ||v_\theta^+(z_{mix}, z_{LR}) - v||^2 + (1-r) ||v_\theta^-(z_{mix}, z_{LR}) - v||^2 \right], \quad (6)$$

其中 $z_{mix} = t_{mid}\epsilon + (1 - t_{mid})z_{LR}$ 、 $v = (z_{mix} - \hat{z}_{HR})/t_{mid}$ 、 $v_\theta^+ = v_\theta$ 、 $v_\theta^- = 2v_{old} - v_\theta$ 以及 r 表示

$\hat{x}_{HR}$ 的奖励。同时, 我们将 $\mathcal{L}_\theta^+$ 定义为正样本优化目标, 将 $\mathcal{L}_\theta^-$ 定义为负样本优化目标:

$$\mathcal{L}_\theta^+ = r ||v_\theta^+ - v||^2, \mathcal{L}_\theta^- = (1-r) ||v_\theta^- - v||^2. \quad (7)$$

特征空间优化。 我们观察到直接使用 $\mathcal{L}_\theta^+$ 和 $\mathcal{L}_\theta^-$ 作为优化目标会导致优化后模型的输出图像出现明显的网格伪影。因此, 我们利用函数 $f(v) = D_\varphi(z_{mix} - t_{mid}v)$, 将速度空间中的优化转换到图像空间。 $v_\theta^+, v_\theta^-, v$ 被输入 $f(v)$ 以获得 $\hat{x}_\theta^+, \hat{x}_\theta^-, \hat{x}_{HR}$ 。然后, 利用SFT训练阶段的 $\mathcal{L}_{rec}$, 我们在图像和特征空间中推导出新的正样本优化目标 $\mathcal{L}_\theta^{' +}$ 和负样本优化目标 $\mathcal{L}_\theta^{' -}$:

$$\mathcal{L}_\theta^{' +} = r \mathcal{L}_{Rec}(\hat{x}_\theta^+, \hat{x}_{HR}), \mathcal{L}_\theta^{' -} = (1-r) \mathcal{L}_{Rec}(\hat{x}_\theta^-, \hat{x}_{HR}). \quad (8)$$

由此, DNF-SR的最终训练目标如下:

$$\mathcal{L}_{NF^2T} = \mathbb{E}_{\pi_{old}(\hat{z}_{HR}|z_{LR}), z_{LR}} \left[r \mathcal{L}_{Rec}(\hat{x}_\theta^+, \hat{x}_{HR}) + (1-r) \mathcal{L}_{Rec}(\hat{x}_\theta^-, \hat{x}_{HR}) \right]. \quad (9)$$

与需要显式概率建模的强化学习方法不同, NF²T通过隐式优化方向直接提升模型性能。这一特性使其特别适合单步超分辨率任务。此外, 与DiffusionDPO相比, NF²T可以利用多个采样图像来确定优化方向, 而DiffusionDPO仅限于成对偏好数据, 这限制了训练过程中的数据利用效率。

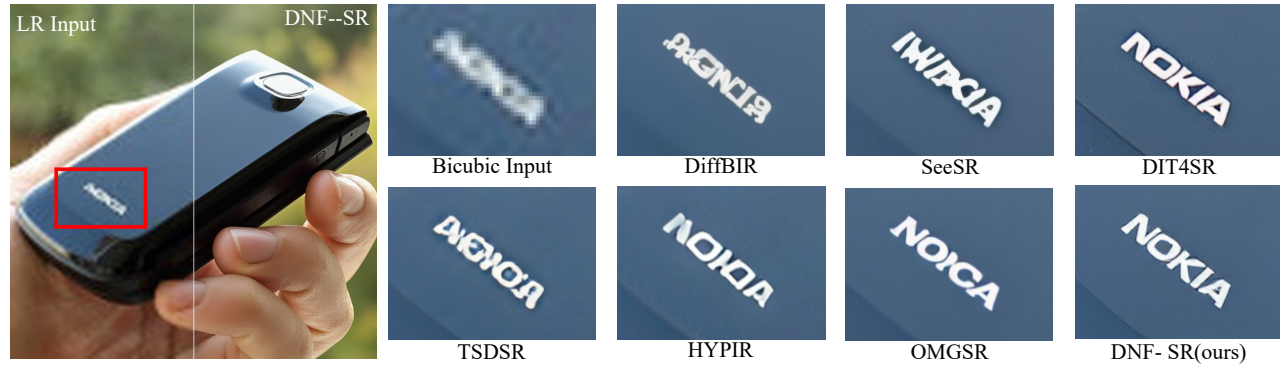
奖励计算。 关于奖励的计算, 对于每次优化步骤中生成的 K 个样本, 我们采用多个评价指标作为奖励, 以确保优化方向的准确性。所使用的评价指标大致分为全参考指标和无参考指标。全参考 (FR) 指标包括 LPIPS [7]和DISTS [11], 而无参考 (NR) 指标包括CLIPQA [42]、MUSIQ[21]和MANIQA [57]。在计算第 i 个指标的 r_i^{raw} 之后, 我们对批次内的奖励进行标准化以获得 r_i^{std} 。假设 r_i^{std} 服从标准高斯分布, 我们通过计算 $r_i = \Phi(X < r_i^{std})$ 得到归一化奖励 r_i , 其中 $\Phi(\cdot)$ 为标准高斯累积分布函数。最后, 将来自不同指标的 r_i 值取平均, 得到最终奖励。

4. 实验

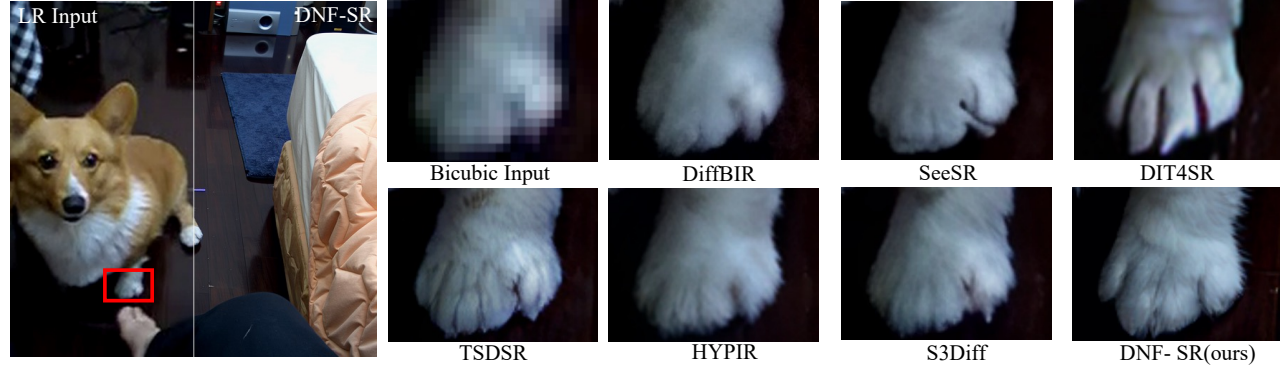
4.1. 实验设置。

训练数据集。 遵循 DiT4SR [13], 我们采用 DIV2K [1], DIV8K [16], Flickr2K [40], NKUSR8K [13],

Caption: A black Nokia flip phone with a camera module and Nokia logo, held in a person's hand, against a blurred green outdoor background.



Caption: A brown and white Corgi dog sits on a dark floor, looking up at the camera, with indoor furniture and a person's foot in the background.



Caption: An aerial view of a parking lot with various cars, surrounded by palm trees and greenery, with modern buildings in the background.

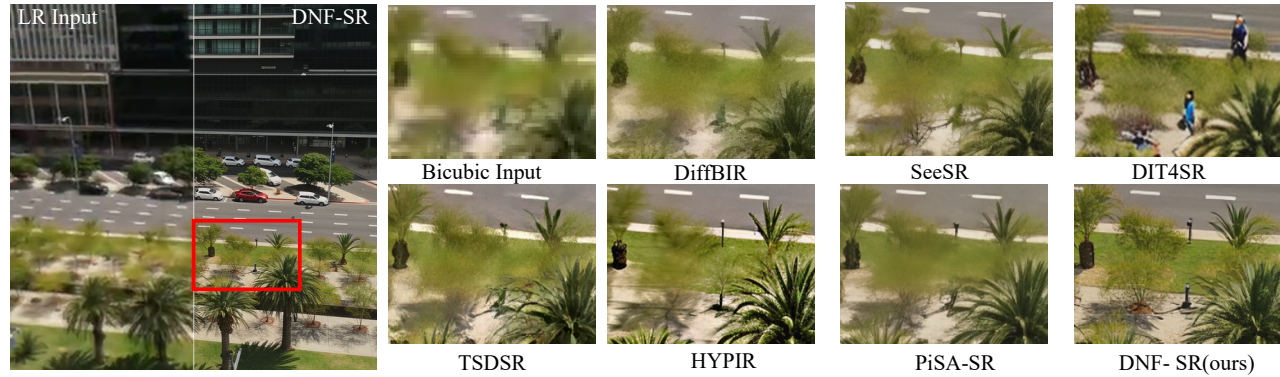


Figure 3. Visual comparisons of different Real-ISR methods. Please zoom in for a better view.

and the first 10K face images from FFHQ [20] as our training dataset. Additionally, we utilize Real-ESRGAN's [43] degradation pipeline to synthesize LR-HR paired data.

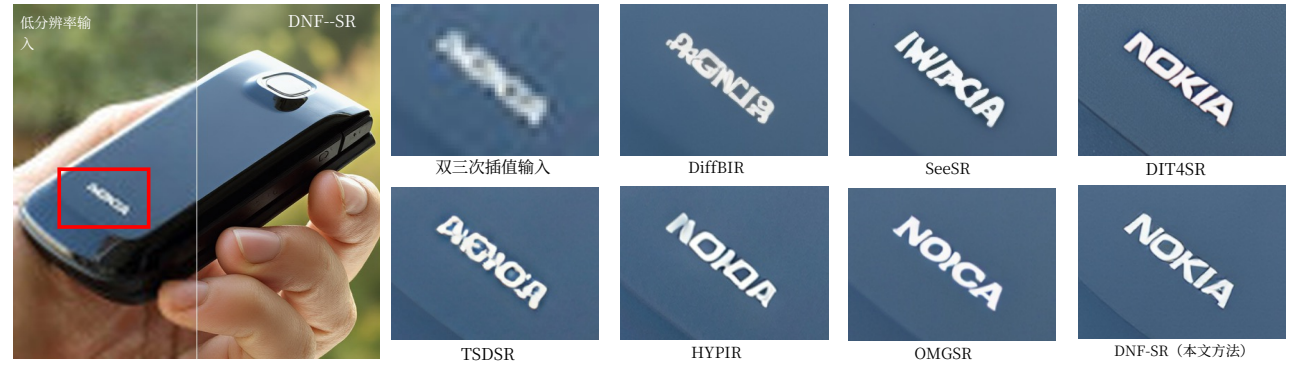
Test Datasets. We evaluate our model using four datasets: RealSR [8], DrealSR [47], DIV2K-Val [1], and RealLQ250 [2]. Among these, DIV2K-Val is a synthetic dataset. RealSR and DrealSR are real-world datasets that include reference HR images, while RealLQ250 is a real-world dataset without reference HR images.

Evaluation Metrics. We employ two reference-based evaluation metrics, PSNR and LPIPS [7], to assess image fidelity and perceptual quality, respectively. However, as

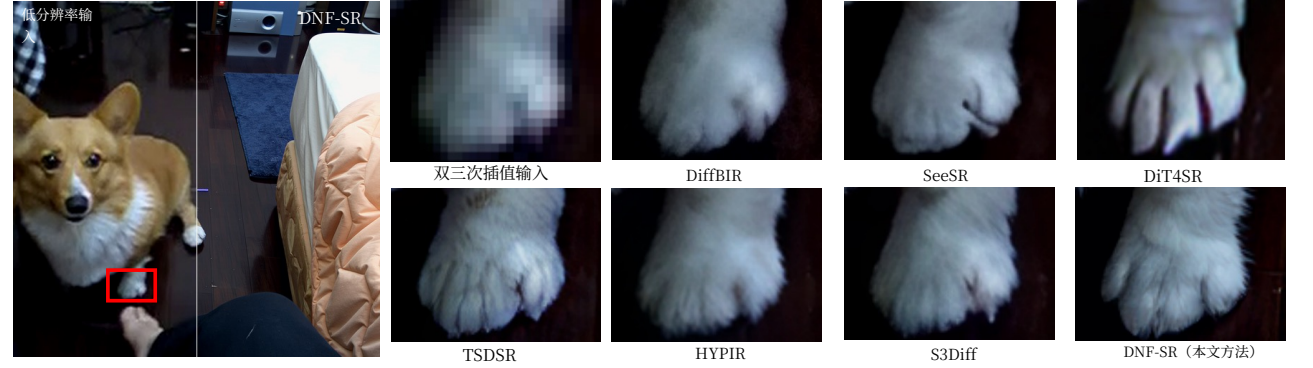
claimed in previous studies [13, 19, 58], full-reference metrics, such as PSNR and SSIM [46], often struggle to accurately reflect the visual effects of restored results. Consequently, we employ several no-reference metrics, including MUSIQ[21], MANIQA [57], CLIPQA [42], QALIGN [48] and VQ-R1 [52], to evaluate image quality.

Implementation Details. We utilize FLUX.1-Kontext-dev [4] as our pre-trained model. During the supervised fine-tuning stage, we fine-tune the VAE Encoder and DiT Block using LoRA [18], while keeping the VAE Decoder fixed. In Eq. (5), we adopt the same settings in OMGSR [53], where $\lambda_1 = 5$, $\lambda_2 = 2$, $\lambda_3 = 5$, $\lambda_4 = 0.5$. We employ

文本描述：一部黑色诺基亚翻盖手机，带有摄像头模组和诺基亚标志，被人握在手中，背景是模糊的绿色户外环境



文本描述：一只棕白色柯基犬坐在深色地板上，抬头看向相机，背景中有室内家具和一个人的脚。



文本描述：航拍视角下的一个停车场，停放着多辆汽车，周围环绕着棕榈树和绿植，背景中有现代建筑

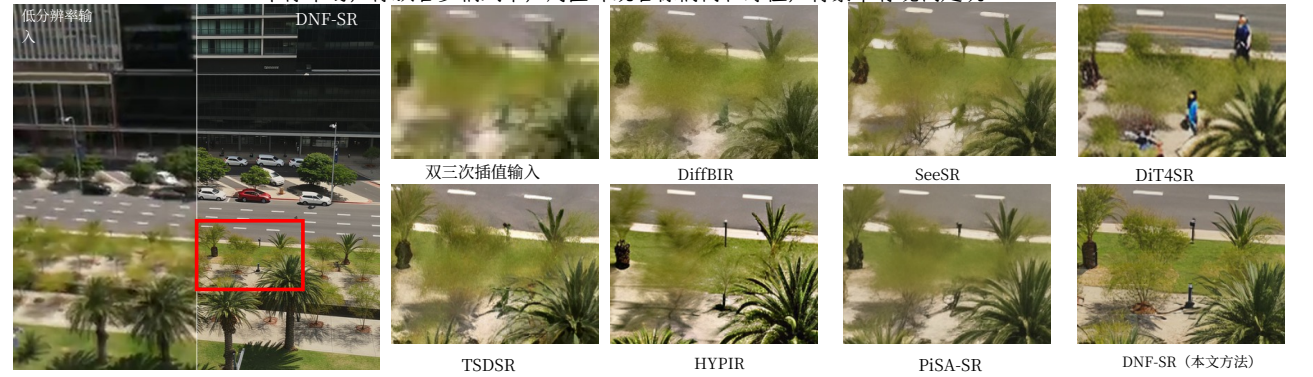


图3. 不同真实图像超分辨率方法的视觉对比。请放大查看以获得更好效果。

以及 FFHQ 的前 10K 张人脸图像 [20] 作为我们的训练数据集。此外，我们利用 Real-ESRGAN 的 [43]退化流水线来合成低分辨率-高分辨率配对数据。

测试数据集。 我们使用四个数据集评估我们的模型：RealSR [8], DrealSR [47], DIV2K-Val [1], 和RealLQ250 [2]。其中，DIV2K-Val是一个合成数据集。RealSR和DrealSR是包含参考高分辨率图像的真实世界数据集，而RealLQ250是一个不含参考高分辨率图像的真实世界数据集。

评估指标。 我们采用两种基于参考的评价指标，峰值信噪比和LPIPS [7]，，分别用于评估图像保真度和感知质量。然而，正如

先前研究所指出的 [13, 19, 58]，，全参考指标（如峰值信噪比和结构相似性 [46]，）往往难以准确反映恢复结果的视觉效果。因此，我们采用多种无参考指标，包括MUSIQ[21]、MANIQA [57]、CLIPQA [42]、QALIGN [48]和VQ-R1 [52]，来评估图像质量。

实现细节。 我们采用 FLUX.1-Kontext-dev [4] 作为预训练模型。在监督微调阶段，我们使用 LoRA [18], 对 VAE 编码器和 DiT 模块进行微调，同时保持 VAE 解码器固定不变。在式 (5) 中，我们采用与 OMGSR 相同的设置[53]，，其中 $\lambda_1 = 5$, $\lambda_2 = 2$, $\lambda_3 = 5$, $\lambda_4 = 0.5$ 。我们采用

Table 1. Quantitative comparison of DNF-SR with SOTA Real-ISR methods on four datasets. DNF-SR(sft) representing DNF-SR only undergoing supervised fine-tuning (without NF^2T post-training) and DNF-SR denoting the full model with NF^2T .

Datasets	Metrics	Multi-step Methods			One-step Methods								
		DiffBIR	SeeSR	DiT4SR	SinSR-1s	OSDiff	S3Diff	PisaSR	TSDSR	HYPiR	OMGSR	DNF-SR(sft)	DNF-SR
RealSR	PSNR↑	24.835	25.147	23.479	26.166	25.141	25.183	25.503	23.404	22.785	25.882	25.628	24.970
	LPIPS↓	0.3650	0.3007	0.3154	0.3062	0.3209	0.2721	0.2672	0.2805	0.3107	0.2779	0.2925	0.3239
	CLIPQA↑	0.7054	0.6706	0.6319	0.6243	0.6613	0.6732	0.6698	0.7196	0.6491	0.6682	0.6903	0.7257
	MUSIQ↑	69.279	69.822	68.197	61.370	67.263	67.828	70.149	70.766	66.559	69.527	70.672	72.040
	MANIQA↑	0.6502	0.6451	0.6594	0.5418	0.6317	0.6424	0.6551	0.6312	0.6558	0.6695	0.6856	0.6930
	QALIGN↑	3.7894	3.7187	3.3974	3.1774	3.6634	3.6632	3.6334	3.7754	3.6931	3.8507	3.9162	4.0718
	VQ-R1↑	3.9388	3.7811	3.7111	3.2470	3.9522	3.9541	3.8162	3.8224	3.9600	4.1285	4.1310	4.2646
DrealSR	PSNR↑	25.904	28.070	25.681	28.149	27.924	27.539	28.319	26.197	25.899	28.928	28.254	28.141
	LPIPS↓	0.4670	0.3174	0.3693	0.3479	0.2966	0.3109	0.2960	0.3115	0.3391	0.2952	0.3210	0.3531
	CLIPQA↑	0.7065	0.6910	0.6658	0.6564	0.6963	0.7133	0.6971	0.7302	0.6429	0.6914	0.6914	0.7559
	MUSIQ↑	66.137	65.085	64.861	57.242	64.692	63.955	66.108	66.120	61.084	65.862	67.247	68.732
	MANIQA↑	0.6221	0.6052	0.6253	0.5027	0.5898	0.6124	0.6160	0.5820	0.6055	0.6315	0.6542	0.6515
	QALIGN↑	3.7142	3.5867	3.3592	3.1929	3.5434	3.6155	3.5828	3.6943	3.5447	3.6877	3.7166	3.7997
	VQ-R1↑	3.6708	3.4905	3.4919	3.1672	3.6619	3.6538	3.5911	3.6615	3.6865	3.8938	3.8478	3.9152
DIV2K	PSNR↑	23.145	23.678	21.789	24.291	23.724	23.530	23.867	22.173	22.252	24.050	23.524	23.631
	LPIPS↓	0.3669	0.3194	0.3490	0.3226	0.2942	0.2581	0.2823	0.2736	0.3042	0.2924	0.3030	0.3234
	CLIPQA↑	0.7300	0.6936	0.6636	0.6532	0.6680	0.7001	0.6927	0.7149	0.6512	0.6828	0.7028	0.7723
	MUSIQ↑	69.872	68.672	68.039	63.276	67.963	67.924	69.679	70.651	65.644	68.624	69.889	71.546
	MANIQA↑	0.6440	0.6222	0.6413	0.5410	0.6131	0.6311	0.6375	0.6077	0.6227	0.6500	0.7009	0.6703
	QALIGN↑	4.1015	3.9766	3.7256	3.5216	3.8352	3.8664	3.8806	3.9271	3.8159	3.9967	4.0604	4.1563
	VQ-R1↑	4.0356	4.0486	3.9834	3.3441	4.0326	3.9861	4.0576	3.8899	3.9462	4.2884	4.2899	4.3630
RealLQ250	CLIPQA↑	0.7137	0.7031	0.7338	0.7140	0.6724	0.7044	0.7055	0.7219	0.6875	0.7435	0.7683	0.7997
	MUSIQ↑	67.531	70.860	72.085	65.305	69.556	69.193	71.245	72.099	68.991	71.975	73.077	73.700
	MANIQA↑	0.5878	0.6001	0.6826	0.5258	0.5782	0.6016	0.6053	0.5829	0.6044	0.6864	0.6990	0.7029
	QALIGN↑	3.9757	4.1543	3.9903	3.7532	4.2481	4.2903	4.2070	4.1687	4.2316	4.3073	4.4017	4.4752
	VQ-R1↑	4.1460	4.3657	4.2968	3.6513	4.4707	4.3840	4.4426	4.2756	4.4300	4.4951	4.5706	4.6090

the AdamW [29] optimizer with a learning rate of 2e-5 and a batch size of 1, training for 5000 steps across 8 H20 GPUs. In the post-training phase, we exclusively fine-tune the DiT Block with LoRA, setting the number of samples per optimization step to 8. For both of these stages, we set the LoRA rank to 64. We employ Qwen3-VL(8B) as an image caption generator to produce captions for both the training-stage high-resolution (HR) images and the inference-stage low-resolution (LR) images. The same caption is also employed during inference for other models.

4.2. Comparison with Existing Methods

Compared Methods. We compare our method with SOTA Diffusion-based Real-ISR methods, which include both single-step and multi-step super-resolution approaches. The multi-step methods comprise DiffBIR [24], SeeSR [50], and DiT4SR [13], while the single-step methods include S3Diff [60], PisaSR [38], SinSR-1s [45], OSEDiff [49], TS-DSR [12], HYPiR [25], and OMGSR [53].

Quantitative Comparisons. Quantitative comparisons with state-of-the-art Real-ISR methods on four benchmarks are presented in Tab. 1. It shows that our DNF-SR achieves overwhelming performance across all no-reference metrics on all benchmarks. Even DNF-SR(sft), which only undergoes supervised fine-tuning without negative-aware feature fine-tuning, still obtains highly competitive performance. Notably, we only use no-reference metrics CLIP-

Table 2. Ablation of the DNF-SR model design. * denotes the use of the text-to-image model Flux as the pre-trained model; all others use Flux-Kontext.

Input	Timestep	LPIPS↓	MUSIQ↑	MANIQA↑	QALIGN↑
z_{mix}^*	t=0.5	0.3349	69.838	0.6729	3.9363
z_{LR}^*	t=0.5	0.3040	70.441	<u>0.6808</u>	3.8771
$(z_{\text{mix}}, z_{\text{LR}})^*$	t=0.5	<u>0.2995</u>	71.116	0.6718	3.9057
$(z_{\text{mix}}, z_{\text{LR}})$	t=0.5	0.2925	<u>70.672</u>	0.6903	<u>3.9162</u>
$(\epsilon, z_{\text{LR}})$	t=1	0.30250	70.333	0.6853	3.8574
$(z_{\text{mix}}, z_{\text{LR}})$	t=0.25	<u>0.2945</u>	70.493	<u>0.6870</u>	3.8688
$(z_{\text{mix}}, z_{\text{LR}})$	t=0.75	0.2960	<u>70.526</u>	0.6790	3.9304
$(z_{\text{mix}}, z_{\text{LR}})$	t=0.5	0.2925	70.672	0.6903	<u>3.9162</u>

IQA, MANIQA, and MUSIQ to calculate the reward, and do not employ the multimodal large language model-based evaluation methods QALIGN and VQ-R1. The state-of-the-art evaluation scores of our method on QALIGN and VQ-R1 fully demonstrate its performance.

Qualitative Comparisons. During inference, we feed the LR image into Qwen3-VL to generate captions, which are then applied to various super-resolution algorithms. The qualitative comparison with other methods is presented in Fig. 3. From the results in the first row, it can be observed that DNF-SR can accurately generate “NOKIA” even when the LR image content is severely corrupted. This is attributed to our improved utilization of the diffusion model’s prior, allowing it to retain strong generative capabilities

表1. DNF-SR与SOTA真实图像超分方法在四个数据集上的定量比较。DNF-SR(sft)表示仅进行监督微调（无 NF^2T 后训练）的 DNF-SR， DNF-SR表示具有 NF^2T 的完整模型。

数据集	评价指标	多步方法			单步方法								
		DiffBIR	SeeSR	DiT4SR	SinSR-1s	OSDiff	S3Diff	PisaSR	TSDSR	HYPIR	OMGSR	DNF-SR(sft)	DNF-SR
RealSR	峰值信噪比↑	24.835	25.147	23.479	26.166	25.141	25.183	25.503	23.404	22.785	25.882	25.628	24.970
	LPIPS↓	0.3650	0.3007	0.3154	0.3062	0.3209	0.2721	0.2672	0.2805	0.3107	0.2779	0.2925	0.3239
	CLIPQA↑	0.7054	0.6706	0.6319	0.6243	0.6613	0.6732	0.6698	0.7196	0.6491	0.6682	0.6903	0.7257
	MUSIQ↑	69.279	69.822	68.197	61.370	67.263	67.828	70.149	70.766	66.559	69.527	70.672	72.040
	MANIQA↑	0.6502	0.6451	0.6594	0.5418	0.6317	0.6424	0.6551	0.6312	0.6558	0.6695	0.6856	0.6930
	QALIGN↑	3.7894	3.7187	3.3974	3.1774	3.6634	3.6632	3.6334	3.7754	3.6931	3.8507	3.9162	4.0718
	VQ-R1↑	3.9388	3.7811	3.7111	3.2470	3.9522	3.9541	3.8162	3.8224	3.9600	4.1285	4.1310	4.2646
DrealSR	峰值信噪比↑	25.904	28.070	25.681	28.149	27.924	27.539	28.319	26.197	25.899	28.928	28.254	28.141
	LPIPS↓	0.4670	0.3174	0.3693	0.3479	0.2966	0.3109	0.2960	0.3115	0.3391	0.2952	0.3210	0.3531
	CLIPQA↑	0.7065	0.6910	0.6658	0.6564	0.6963	0.7133	0.6971	0.7302	0.6429	0.6914	0.6914	0.7559
	MUSIQ↑	66.137	65.085	64.861	57.242	64.692	63.955	66.108	66.120	61.084	65.862	67.247	68.732
	MANIQA↑	0.6221	0.6052	0.6253	0.5027	0.5898	0.6124	0.6160	0.5820	0.6055	0.6315	0.6542	0.6515
	QALIGN↑	3.7142	3.5867	3.3592	3.1929	3.5434	3.6155	3.5828	3.6943	3.5447	3.6877	3.7166	3.7997
	VQ-R1↑	3.6708	3.4905	3.4919	3.1672	3.6619	3.6538	3.5911	3.6615	3.6865	3.8938	3.8478	3.9152
DIV2K	峰值信噪比↑	23.145	23.678	21.789	24.291	23.724	23.530	23.867	22.173	22.252	24.050	23.524	23.631
	LPIPS↓	0.3669	0.3194	0.3490	0.3226	0.2942	0.2581	0.2823	0.2736	0.3042	0.2924	0.3030	0.3234
	CLIPQA↑	0.7300	0.6936	0.6636	0.6532	0.6680	0.7001	0.6927	0.7149	0.6512	0.6828	0.7028	0.7723
	MUSIQ↑	69.872	68.672	68.039	63.276	67.963	67.924	69.679	70.651	65.644	68.624	69.889	71.546
	MANIQA↑	0.6440	0.6222	0.6413	0.5410	0.6131	0.6311	0.6375	0.6077	0.6227	0.6500	0.7009	0.6703
	QALIGN↑	4.1015	3.9766	3.7256	3.5216	3.8352	3.8664	3.8806	3.9271	3.8159	3.9967	4.0604	4.1563
	VQ-R1↑	4.0356	4.0486	3.9834	3.3441	4.0326	3.9861	4.0576	3.8899	3.9462	4.2884	4.2899	4.3630
RealLQ250	CLIPQA↑	0.7137	0.7031	0.7338	0.7140	0.6724	0.7044	0.7055	0.7219	0.6875	0.7435	0.7683	0.7997
	MUSIQ↑	67.531	70.860	72.085	65.305	69.556	69.193	71.245	72.099	68.991	71.975	73.077	73.700
	MANIQA↑	0.5878	0.6001	0.6826	0.5258	0.5782	0.6016	0.6053	0.5829	0.6044	0.6864	0.6990	0.7029
	QALIGN↑	3.9757	4.1543	3.9903	3.7532	4.2481	4.2903	4.2070	4.1687	4.2316	4.3073	4.4017	4.4752
	VQ-R1↑	4.1460	4.3657	4.2968	3.6513	4.4707	4.3840	4.4426	4.2756	4.4300	4.4951	4.5706	4.6090

AdamW [29]优化器，学习率设为 2e-5，批大小为 1，在 8 个 H20 GPU 上训练 5000 步。在后训练阶段，我们仅对 DiT 模块进行 LoRA 微调，将每个优化步骤的样本数设为 8。在这两个阶段中，我们均将 LoRA 秩设为 64。我们采用 Qwen3-VL(8B) 作为图像描述生成器，为训练阶段的高分辨率（HR）图像和推理阶段的低分辨率（LR）图像生成文本描述。在其他模型的推理过程中，也采用相同的文本描述。

4.2. 与现有方法的比较

比较方法。我们将我们的方法与最先进的基于扩散的真实图像超分辨率方法进行比较，这些方法包括单步和多步超分辨率方法。多步方法包括DiffBIR [24]、SeeSR [50]和DiT4SR [13]，而单步方法包括S3Diff [60]、PisaSR [38]、SinSR-1s [45]、OSEDiff [49]、TS-DSR [12]、HYPiR [25]和OMGSR [53]。**定量比较。**与最先进的真实图像超分辨率方法在四个基准上的定量比较见表1。结果表明，我们的 DNF-SR 在所有基准上的所有无参考指标上都取得了压倒性的性能。即使仅进行监督微调而未进行负样本感知特征微调的 DNF-SR(sft)，也仍然获得了极具竞争力的性能。值得注意的是，我们仅使用无参考指标 CLIP-

表2. DNF-SR模型设计的消融实验。* 表示使用文生图模型 Flux作为预训练模型；其余均使用Flux-Kontext。

输入	时间步	LPIPS↓	MUSIQ↑	MANIQA↑	QALIGN↑
z_{mix}^*	t=0.5	0.3349	69.838	0.6729	3.9363
z_{LR}^*	t=0.5	0.3040	70.441	<u>0.6808</u>	3.8771
$(z_{\text{mix}}, z_{\text{LR}})^*$	t=0.5	<u>0.2995</u>	71.116	0.6718	3.9057
$(z_{\text{mix}}, z_{\text{LR}})$	t=0.5	0.2925	<u>70.672</u>	0.6903	<u>3.9162</u>
$(\epsilon, z_{\text{LR}})$	t=1	0.30250	70.333	0.6853	3.8574
$(z_{\text{mix}}, z_{\text{LR}})$	t=0.25	<u>0.2945</u>	70.493	<u>0.6870</u>	3.8688
$(z_{\text{mix}}, z_{\text{LR}})$	t=0.75	0.2960	<u>70.526</u>	0.6790	3.9304
$(z_{\text{mix}}, z_{\text{LR}})$	t=0.5	0.2925	70.672	0.6903	<u>3.9162</u>

IQA、MANIQA 和 MUSIQ 来计算奖励，并未采用基于多模态大语言模型的评估方法 QALIGN 和 VQ-R1。我们的方法在 QALIGN 和 VQ-R1 上取得的最先进评估分数充分证明了其性能。

Caption: Stacked Siemens product boxes with gray and beige design, featuring brand logo and leaf motif, arranged in orderly rows. Make this image more realistic and detailed.

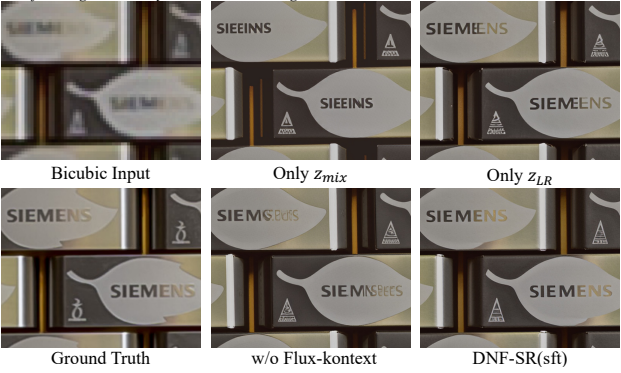


Figure 4. Visualizing the ablation study of the model design.

when applied to super-resolution tasks. It can be seen that DiT4SR also generates a relatively good logo, but due to its multi-step super-resolution approach, DNF-SR has an advantage in performance. The results from the second row indicate that our method can produce more realistic outcomes (e.g., the dog’s toes and fur). The third row demonstrates that our method can generate clearer results.

4.3. Ablation Study

To further demonstrate the effectiveness of DNF-SR, we conduct an ablation study on RealSR. Specifically, we perform ablation experiments on the **model design** and post-training algorithm NF²T used in DNF-SR, respectively.

Model Design. In Tab. 2, we conduct ablation experiments on three aspects of the DNF-SR model structure: dual-path input, using the editing model, and denoising at intermediate timesteps. 1) **dual-path input and using editing model.** Tab. 2 shows that the dual-input in DNF-SR achieves better performance compared to using only LR latent z_{LR} or only noisy LR latent z_{mix} . Additionally, using the image editing model FLUX-Kontext further enhances performance. As depicted in Fig. 5, DNF-SR effectively leverages the prior knowledge of the generative model under a given caption, producing more accurate content while maintaining fidelity. 2) **denoising at intermediate timesteps.** We also report the metrics for single-step denoising of DNF-SR at different timesteps t . Compared to denoising from pure noise in a single step, denoising at any intermediate timestep yields better performance. To balance realism and fidelity, we chose $t = 0.5$.

Negative-aware Feature Fine-Tuning. In Tab. 3, we conduct ablation experiments on the post-training algorithm of our proposed NF²T. It can be observed that DNF-SR achieves significant improvements in all no-reference metrics, especially QALIGN (not involved in reward calculation). As shown in Fig. 5, DNF-SR with NF²T yields notably clearer results. In contrast, the HR ground truth is

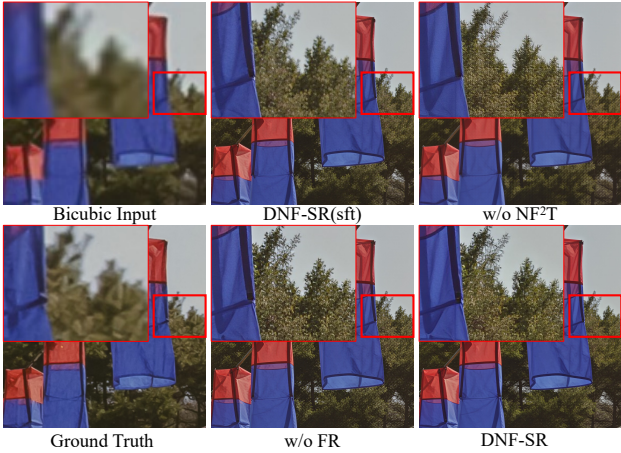


Figure 5. Visualizing the ablation study of the NF^2T method.

blurrier, which leads to a decrease in the reference-based LPIPS metric for DNF-SR. Meanwhile, as shown in Fig. 5, if NFT is used for optimization in the latent space, it exhibits obvious grid artifacts. Regarding the use of the reward model, we find that incorporating full-reference (FR) metrics when using NF2T can further improve no-reference (NR) metrics while preventing degradation of FR metrics. Additionally, it can generate more natural results.

Table 3. Ablation study of the NF²T method.

Setting	Reward Model	LPIPS↓	MUSIQ↑	MANIQA↑	QALIGN↑
-	-	0.2925	70.6717	0.6903	3.9162
NFT	NR+FR	0.3250	71.1682	0.6413	3.943
NF ² T	NR	0.3255	71.0911	0.6925	3.9979
NF ² T	NR+FR	0.3239	72.0396	0.693	4.0718

5. Conclusion

This paper presents DNF-SR, a novel framework for real-world image super-resolution (Real-ISR). DNF employs a dual-input architecture that concatenates noisy LR latents with original LR latents to narrow the distribution gap between LR inputs and diffusion models’ native inputs while preserving content fidelity. Additionally, the Flux-Kontext image-editing pretrained model is utilized to better leverage generative model priors. Furthermore, we propose Negative-aware Feature Fine-Tuning (NF²T), which shifts optimization from the latent space to image and feature spaces, using aggregated rewards to enhance realism and suppress artifacts. Experiments demonstrate that DNF-SR achieves state-of-the-art performance across multiple benchmarks and can generate superior results.

文本描述：堆叠的西门子产品包装盒，采用灰色和米色设计，带有品牌标志和叶片图案，排列整齐有序。让这张图像更逼真、更细腻。

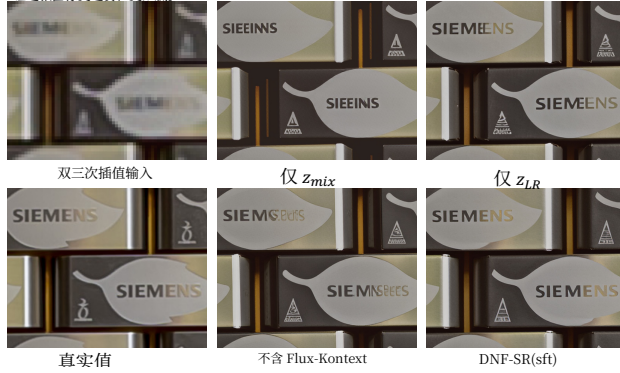


图4. 模型设计消融研究的可视化。

可以看出，DiT4SR也生成了相对不错的标志，但由于其多步超分辨率方法，DNF-SR在性能上具有优势。第二行的结果表明，我们的方法能产生更真实的结果（例如狗的脚趾和毛发）。第三行表明，我们的方法能生成更清晰的结果。

4.3. 消融研究

为了进一步证明DNF-SR的有效性，我们在RealSR上进行了消融研究。具体来说，我们分别对DNF-SR中使用的**模型设计**和后训练算法NF²T进行了消融实验。

模型设计。在表2中，我们对DNF-SR模型结构的三个方面进行了消融实验：双路径输入、使用编辑模型以及在中间时间步去噪。1) **双路径输入和使用编辑模型。**表2显示，与仅使用低分辨率潜变量 z_{LR} 或仅使用带噪低分辨率潜变量 z_{mix} 相比，DNF-SR中的双输入取得了更好的性能。此外，使用图像编辑模型FLUX-Kontext进一步提升了性能。如图5所示，DNF-SR在给定文本描述的情况下有效利用了生成模型的先验知识，在保持保真度的同时生成了更准确的内容。2) **在中间时间步去噪。**我们还报告了DNF-SR在不同时间步进行单步去噪的评价指标 t 。与从纯噪声进行单步去噪相比，在任意中间时间步进行去噪都能获得更好的性能。为了平衡真实感和保真度，我们选择了 $t = 0.5$ 。

负样本感知特征微调。在表3中，我们对所提出的NF²T后训练算法进行了消融实验。可以观察到，DNF-SR在所有无参考指标上都取得了显著提升，尤其是QALIGN（未参与奖励计算）。如图5所示，采用NF²T的DNF-SR得到了明显更清晰的结果。相比之下，高分辨率真值则

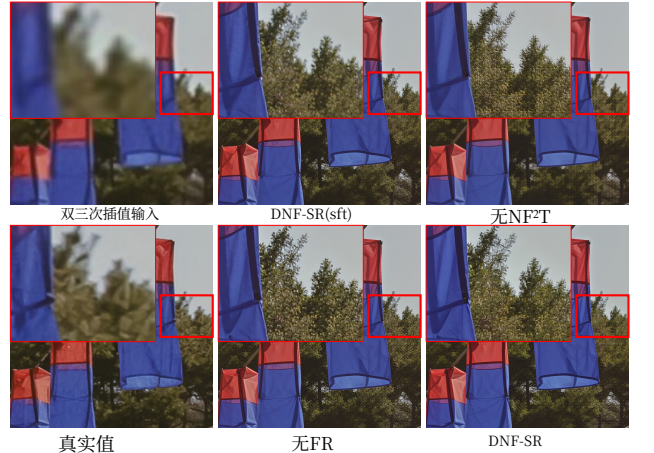


图5. NF^2T 方法消融研究的可视化。

更模糊，这导致DNF-SR的基于参考的LPIPS指标有所下降。同时，如图5所示，如果在潜空间中使用NFT进行优化，则会出现明显的网格伪影。关于奖励模型的使用，我们发现，在使用NF2T时引入全参考（FR）指标可以进一步提升无参考（NR）指标，同时防止FR指标退化。此外，它还能生成更自然的结果。

表3. NF²T方法的消融研究。

设置	奖励模型	LPIPS↓	MUSIQ↑	MANIQA↑	QALIGN↑
-	-	0.2925	70.6717	0.6903	3.9162
NFT	NR+FR	0.3250	71.1682	0.6413	3.943
NF ² T	NR	0.3255	71.0911	0.6925	3.9979
NF ² T	NR+FR	0.3239	72.0396	0.693	4.0718

5. 结论

本文提出了DNF-SR，一种用于真实世界图像超分辨率（Real-ISR）的新型框架。DNF采用双输入架构，将带噪低分辨率潜变量与原始低分辨率潜变量拼接起来，以缩小低分辨率输入与扩散模型原生输入之间的分布差距，同时保持内容保真度。此外，利用Flux-Kontext图像编辑预训练模型来更好地利用生成模型先验。进一步地，我们提出了负向感知特征微调（NF²T），将优化从潜空间转移到图像与特征空间，利用聚合奖励来增强真实感并抑制伪影。实验表明，DNF-SR在多个基准上达到了最先进的性能，并能生成更优的结果。

6. Acknowledgments

This work was supported in part by the National Natural Science Foundation of China (62306153, 62225604), Tianjin Natural Science Foundation Project (25ZXRGGX00290, 24JCQJC00020, 25JCQNJ01390), the Young Elite Scientists Sponsorship Program by CAST (YESS20240686), the Fundamental Research Funds for the Central Universities (Nankai University, 63253223, 63253219), “Science and Technology Yongjiang 203” key technology breakthrough plan project (2024Z120), Chinese government-guided local science and technology development fund projects (scientific and technological achievement transfer and transformation projects)(254Z0102G) and Shenzhen Science and Technology Program (JCYJ20240813114237048). The computational devices is supported by the Supercomputing Center of Nankai University (NKSC).

6. 致谢

本工作部分由国家自然科学基金（62306153、62225604）、天津市自然科学基金项目（25ZXRGGX00290、24JCQJC00020、25JCQNJ01390）、中国科协青年人才托举工程（YESS20240686）、中央高校基本科研业务费（南开大学，63253223、63253219）、“科技创新甬江203”关键技术突破计划项目（2024Z120）、中国政府引导地方科技发展资金项目（科技成果转化项目）（254Z0102G）以及深圳市科技计划（JCYJ20240813114237048）资助。计算设备由南开大学超级计算中心（NKSC）支持。

Supplementary Material

In this Supplementary Material, we provide additional details, including the comparison with GAN-based methods in Section A, discussion on perception and restoration in Section B, more visual comparisons in Section C, and the algorithm in Section D. We conduct these additional comparisons and analyses to validate the effectiveness of DNF-SR.

A. Comparison with GAN-based Methods

We compare DNF-SR with three GAN-based Real-ISR methods: BSRGAN [61], RealESRGAN [43], and LDL [23]. Quantitative evaluations are conducted on the DIV2K [1], RealSR [8], DrealSR [47] and RealLQ [2] datasets, with results summarized in Tab. 4. The experimental results demonstrate that DNF-SR, leveraging a dual-input strategy and a novel post-training optimization method NF²T, achieves significantly superior no-reference metrics compared to GAN-based methods.

Additionally, Fig 7 presents a visual comparison between DNF-SR and other GAN-based methods. The results show that DNF-SR reconstructs more photorealistic and natural outcomes. When compared to GAN-based methods, DNF-SR demonstrates distinct advantages in visual fidelity. Specifically, it achieves higher precision in restoring structured elements (e.g., text and architectural details) while rendering complex materials such as fabrics and natural textures with enhanced realism. This enables DNF-SR to more accurately reproduce the fine-grained detail hierarchy and authentic visual texture characteristic of high-resolution images, outperforming the GAN-based methods in both structural integrity and perceptual quality.

B. Discussion on Perception and Restoration

In experiments, we observe that existing multi-modal large language models (MLLMs) can effectively perceive the content in images. Even for challenging low-resolution (LR) images, they can infer reasonable content for blurred regions based on the overall image information. Meanwhile, current DiT-based generative models can adhere well to captions for image generation. However, in SR tasks, restoring strongly semantic structures such as text and logos is extremely challenging. As shown in the first row of Fig. 6, when no additional caption is used for the SR model, it is difficult to restore text with normal semantics. As shown in Fig 6, when captions generated by MLLMs are used as conditions, DiT-based SR models can effectively alleviate this issue. Nevertheless, when we manually replace “Nokia” with “Nckia”, DiT4SR exhibits poor prompt-following performance. This is because when applying text-to-image models to SR task, the model is caused to focus more on LR images, which impairs the inherent prompt-following

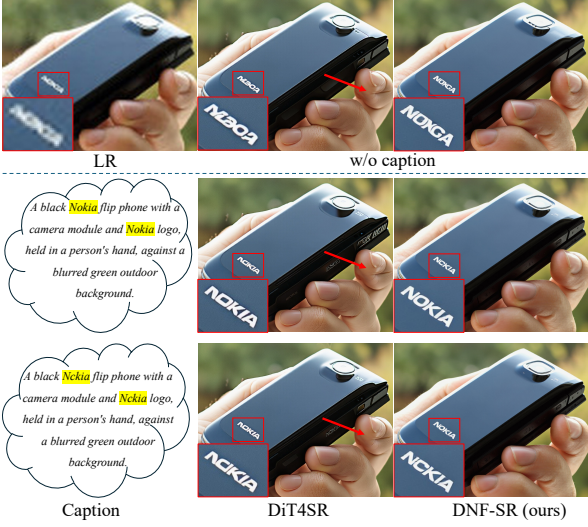


Figure 6. Visual comparisons between DNF-SR and DiT4SR when using different captions including no caption, a reasonable caption with “Nokia”, and an unreasonable caption where “Nokia” is replaced with “Nckia”. DiT4SR exhibits structural issues at the position indicated by the red arrow.

capability of the original text-to-image model. In contrast, our method DNF-SR narrows the gap between LR and the original input of generative models through a dual-path input design. It also initializes with an image editing model to better perceive the information of LR used as conditions. This enables DNF-SR to better preserve the inherent generation capability and prompt-following ability of the original generative model when applied to SR tasks, thereby allowing it to better adhere to captions and restore more realistic and reasonable images.

However, current SR methods still obtain captions by leveraging MLLMs to perceive image content before applying them to restoration tasks. Only using text caption may sometimes fail to fully convey image information, and the separate execution of perception and restoration steps also introduces redundancy. Thus, it is highly meaningful to develop an integrated perception-restoration model that can accurately perceive semantic information in LR images and perform restoration within a single framework.

C. More Visual Comparisons

In Fig. 8 and 9, we provide more visual comparisons with other diffusion-based Real-ISR methods. As shown in Fig. 8, DNF-SR can better adhere to the content of the caption and restore more accurate text information. And in close-up scenarios, DNF-SR can better restore the texture and details of the image. Meanwhile, as shown in Fig. 9, DNF-SR can restore more realistic images under se-

补充材料

在本补充材料中，我们提供了更多细节，包括第A节中与基于GAN的方法的对比、第B节中关于感知与恢复的讨论、第C节中更多的视觉对比，以及第D节中的算法。我们进行了这些额外的对比与分析，以验证 DNF-SR的有效性。

A. 与基于GAN的方法的对比

我们将DNF-SR与三种基于GAN的真实图像超分辨率方法进行对比：BSRGAN [61], RealESRGAN [43], 和 LDL[23]。定量评估在DIV2K[1], RealSR [8], DrealSR [47]和RealLQ [2] 数据集上进行，结果汇总于表4。实验结果表明，DNF-SR凭借双输入策略和新颖的训练后优化方法NF²T，在无参考指标上显著优于基于GAN的方法。

此外，图7展示了DNF-SR与其他基于GAN的方法之间的视觉对比。结果表明，DNF-SR重建出的结果更加照片级真实和自然。与基于GAN的方法相比，DNF-SR在视觉保真度方面展现出明显优势。具体而言，它在恢复结构化元素（如文本和建筑细节）方面达到了更高精度，同时以更强的真实感渲染织物和自然纹理等复杂材质。这使得DNF-SR能够更准确地再现高分辨率图像特有的细粒度细节层次和真实视觉纹理，在结构完整性和感知质量两方面均优于基于GAN的方法。

B. 关于感知与恢复的讨论

在实验中，我们观察到现有的多模态大语言模型（MLLMs）能够有效感知图像中的内容。即使对于具有挑战性的低分辨率（LR）图像，它们也能基于整体图像信息为模糊区域推断出合理的内容。同时，当前基于DiT的生成模型能够很好地遵循文本描述进行图像生成。然而，在超分辨率任务中，恢复诸如文本和标志等强语义结构极具挑战性。如图6第一行所示，当超分辨率模型未使用额外的文本描述时，很难恢复具有正常语义的文本。如图6所示，当使用多模态大语言模型生成的文本描述作为条件时，基于DiT的超分辨率模型能够有效缓解这一问题

问题。然而，当我们手动将“Nokia”替换为“Nckia”时，DiT4SR表现出较差的提示遵循性能。这是因为当将文生图模型应用于超分辨率任务时，模型会被引导更多地关注低分辨率图像，从而损害了原始文生图模型固有的提示遵循能力。

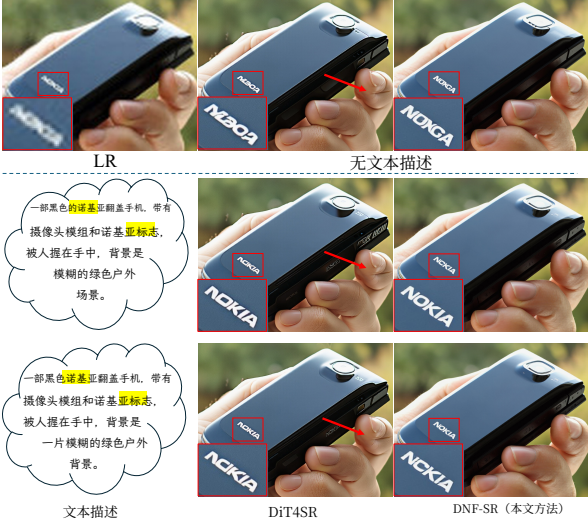


图6. DNF-SR与DiT4SR在使用不同文本描述时的视觉对比，包括无文本描述、包含“Nokia”的合理文本描述，以及将“Nokia”替换为“Nckia”的不合理文本描述。DiT4SR在红色箭头所指位置出现了结构问题。

相比之下，我们的方法DNF-SR通过双路径输入设计缩小了低分辨率与生成模型原始输入之间的差距。它还以图像编辑模型进行初始化，以更好地感知用作条件的低分辨率信息。这使得DNF-SR在应用于超分辨率任务时，能够更好地保留原始生成模型固有的生成能力和提示遵循能力，从而使其更好地遵循文本描述，并恢复出更真实、更合理的图像。

然而，当前的超分辨率方法仍依赖多模态大语言模型先感知图像内容，再将其应用于恢复任务。仅使用文本描述有时无法充分传达图像信息，且感知与恢复步骤分离执行也会引入冗余。因此，开发一种感知-恢复一体化模型极具意义，该模型能够在单一框架内准确感知低分辨率图像中的语义信息并完成恢复。

C. 更多视觉对比

在图8和9中，我们提供了与其他基于扩散的真实图像超分辨率方法的更多视觉对比。如图8所示，DNF-SR能够更好地遵循文本描述的内容，并恢复出更准确的文本信息。在特写场景中，DNF-SR能够更好地恢复图像的纹理和细节。同时，如图9所示，DNF-SR能够在严重降

Table 4. A comprehensive evaluation against state-of-the-art GAN-based methods across synthetic and real-world datasets. The top-performing results under each metric are marked in **red**.

Datasets	Methods	PSNR $\uparrow$	LPIPS [7] $\downarrow$	CLIPQA [42] $\uparrow$	MUSIQ [21] $\uparrow$	MAINQA [57] $\uparrow$	QALIGN [48] $\uparrow$	VQ-R1 [52] $\uparrow$
<i>DIV2k</i>	BSRGAN	24.583	0.3351	0.5246	61.193	0.5040	3.1703	3.3063
	RealESRGAN	24.293	0.3112	0.5276	61.049	0.5484	3.2764	3.2623
	LDL	23.828	0.3256	0.5179	60.040	0.5328	3.1798	3.1018
	DNF-SR	23.631	0.3234	0.7723	71.546	0.6703	4.1563	4.3630
<i>DrealSR</i>	BSRGAN	28.702	0.2858	0.5092	57.159	0.4844	2.9572	3.0559
	RealESRGAN	28.618	0.2818	0.4517	54.275	0.4902	2.8638	2.7683
	LDL	28.196	0.2790	0.4473	53.948	0.4894	2.8576	2.6129
	DNF-SR	28.141	0.3531	0.7559	68.732	0.6515	3.7997	3.9152
<i>RealSR</i>	BSRGAN	26.379	0.2656	0.5114	63.283	0.5419	3.1829	3.4907
	RealESRGAN	25.687	0.2710	0.4489	60.364	0.5504	3.1081	3.1342
	LDL	25.280	0.2750	0.4556	60.930	0.5495	3.0898	2.9897
	DNF-SR	24.970	0.3239	0.7257	70.040	0.6930	4.0718	4.2646
<i>RealLQ250</i>	BSRGAN	-	-	0.5940	66.289	0.5963	3.4794	3.8124
	RealESRGAN	-	-	0.6253	66.990	0.6148	3.6471	3.8088
	LDL	-	-	0.6183	67.027	0.6147	3.6357	3.6940
	DNF-SR	-	-	0.7997	73.700	0.7029	4.4752	4.6090

vere degradation. These examples all demonstrate the performance and robustness of DNF-SR for Real-ISR.

D. Algorithm Details

The training of DNF-SR consists of two stages: supervised fine-tuning (SFT) and post-training. In the SFT stage, we use multiple losses to perform supervised fine-tuning on the paired (x_L, x_H, c) dataset. In the post-training stage, we adopt a Negative-aware Feature Fine-Tuning method for reinforcement learning. Specifically, we sample K noises to generate K restored images, then compute rewards using multiple reward functions, which are normalized and aggregated into a single r . Subsequently, we define positive and negative optimization directions to improve model performance. Here, $K = 8$. Details are in Algorithm 1.

References

[1] Eirikur Agustsson and Radu Timofte. Ntire 2017 challenge on single image super-resolution: Dataset and study. In *CVPRW*, 2017.

[2] Yuang Ai, Xiaoqiang Zhou, Huaibo Huang, Xiaotian Han, Zhengyu Chen, Quanzeng You, and Hongxia Yang. Dream-clear: High-capacity real-world image restoration with privacy-safe dataset curation. In *NeurIPS*, 2025.

[3] Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *ArXiv preprint*, 2022.

[4] Stephen Batifol, Andreas Blattmann, Frederic Boesel, Saksham Consul, Cyril Diagne, Tim Dockhorn, Jack English, Zion English, Patrick Esser, Sumith Kulal, et al. Flux. 1 kontext: Flow matching for in-context image generation and editing in latent space. *ArXiv preprint*, 2025.

[5] Kevin Black, Michael Janner, Yilun Du, Ilya Kostrikov, and Sergey Levine. Training diffusion models with reinforcement learning. *ArXiv preprint*, 2023.

[6] blackforestlabs.ai. Flux, offering state-of-the-art performance image generation, 2024.

[7] Yochai Blau and Tomer Michaeli. The perception-distortion tradeoff. In *CVPR*, 2018.

[8] Jianrui Cai, Hui Zeng, Hongwei Yong, Zisheng Cao, and Lei Zhang. Toward real-world single image super-resolution: A new benchmark and a new model. In *ICCV*, 2019.

[9] Xiaofeng Cao, Mingwei Xu, Xin Yu, Jiangchao Yao, Wei Ye, Shengjun Huang, Minling Zhang, Ivor Tsang, Yew-Soon Ong, James T Kwok, et al. Analytical survey of learning with low-resource data: From analysis to investigation. *ACM Computing Surveys*, 2025.

[10] Junsong Chen, Jincheng Yu, Chongjian Ge, Lewei Yao, Enze Xie, Yue Wu, Zhongdao Wang, James Kwok, Ping Luo, Huchuan Lu, et al. Pixart-*alpha*: Fast training of diffusion transformer for photorealistic text-to-image synthesis. *ArXiv preprint*, 2023.

[11] Keyan Ding, Kede Ma, Shiqi Wang, and Eero P Simoncelli. Image quality assessment: Unifying structure and texture similarity. *TPAMI*, 2020.

[12] Linwei Dong, Qingnan Fan, Yihong Guo, Zhonghao Wang, Qi Zhang, Jinwei Chen, Yawei Luo, and Changqing Zou. Tsd-sr: One-step diffusion with target score distillation for real-world image super-resolution. In *CVPR*, 2025.

[13] Zheng-Peng Duan, Jiawei Zhang, Xin Jin, Ziheng Zhang, Zheng Xiong, Dongqing Zou, Jimmy S Ren, Chunle Guo, and Chongyi Li. Dit4sr: Taming diffusion transformer for real-world image super-resolution. In *ICCV*, 2025.

[14] Ying Fan, Olivia Watkins, Yuqing Du, Hao Liu, Moonkyung Ryu, Craig Boutilier, Pieter Abbeel, Mohammad Ghavamzadeh, Kangwook Lee, and Kimin Lee. Dpok: Reinforcement learning for fine-tuning text-to-image diffusion models. In *NeurIPS*, 2023.

表4. 在合成数据集和真实世界数据集上针对最先进的基于GAN的方法进行的综合评估。每个评价指标下表现最佳的结果以**红色**标出。

数据集	方法	PSNR $\uparrow$	LPIPS [7] $\downarrow$	CLIPQA [42] $\uparrow$	MUSIQ [21] $\uparrow$	MAINQA [57] $\uparrow$	QALIGN [48] $\uparrow$	VQ-R1 [52] $\uparrow$
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	LDL	-	-	0.6183	67.027	0.6147	3.6357	3.6940
	DNF-SR	-	-	0.7997	73.700	0.7029	4.4752	4.6090

质情况下恢复出更真实感的图像。这些例子都证明了 DNF-SR 在真实图像超分辨率中的性能和鲁棒性。

D. 算法细节

DNF-SR 的训练包括两个阶段：监督微调（SFT）和后训练。在 SFT 阶段，我们使用多种损失在配对 (x_L, x_H, c) 数据集上进行监督微调。在后训练阶段，我们采用负向感知特征微调方法进行强化学习。具体来说，我们采样 K 噪声以生成 K 恢复图像，然后使用多个奖励函数计算奖励，这些奖励被归一化并聚合为一个 r 。随后，我们定义正优化方向和负优化方向来提升模型性能。这里， $K = 8$ 。细节见算法 1。

参考文献

[1] Eirikur Agustsson 和 Radu Timofte. Ntire 2017 挑战赛：单图像超分辨率数据集与研究。载于*CVPRW*，2017 年。[2] Yuang Ai、Xiaoqiang Zhou、Huaibo Huang、Xiaotian Han、Zhengyu Chen、Quanzeng You 和 Hongxia Yang。Dream-clear：具有隐私安全数据集构建的高容量真实世界图像复原。载于*NeurIPS*，2025 年。[3] Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan 等。基于人类反馈的强化学习训练有用且无害的助手。*arXiv* 预印本，2022 年。[4] Stephen Batifol, Andreas Blattmann, Frederic Boesel, Saksham Consul, Cyril Diagne, Tim Dockhorn, Jack English, Zion English, Patrick Esser, Sumith Kulal 等。Flux.1 kontext：潜空间中的上下文图像生成与编辑流匹配。*arXiv* 预印本，2025 年。

[5] Kevin Black, Michael Janner, Yilun Du, Ilya Kostrikov 和 Sergey Levine。使用强化学习训练扩散模型。*arXiv* 预印本，2023 年。[6] blackforestlabs.ai。Flux，提供最先进性能的图像生成，2024 年。[7] Yochai Blau 和 Tomer Michaeli。感知-失真权衡。载于*CVPR*，2018 年。[8] Jianrui Cai, Hui Zeng, Hongwei Yong, Zisheng Cao 和 Lei Zhang。面向真实世界单图像超分辨率：新基准与新模型。载于*ICCV*，2019 年。[9] Xiaofeng Cao, Mingwei Xu, Xin Yu, Jiangchao Yao, Wei Ye, Shengjun Huang, Minling Zhang, Ivor Tsang, Yew-Soon Ong, James T Kwok 等。低资源数据学习的分析性综述：从分析到研究。*ACM Computing Surveys*，2025 年。[10] Junsong Chen, Jincheng Yu, Chongjian Ge, Lewei Yao, Enze Xie, Yue Wu, Zhongdao Wang, James Kwok, Ping Luo, Huchuan Lu 等。Pixart-*alpha*：面向照片级真实感文本到图像合成的扩散Transformer快速训练。*arXiv* 预印本，2023 年。[11] Keyan Ding, Kede Ma, Shiqi Wang 和 Eero P Simoncelli。图像质量评估：统一结构与纹理相似性。*TPAMI*，2020 年。[12] Linwei Dong, Qingnan Fan, Yihong Guo, Zhonghao Wang, Qi Zhang, Jinwei Chen, Yawei Luo 和 Changqing Zou。Tsd-sr：面向真实世界图像超分辨率的单步目标得分蒸馏扩散。载于*CVPR*，2025 年。[13] Zheng-Peng Duan, Jiawei Zhang, Xin Jin, Ziheng Zhang, Zheng Xiong, Dongqing Zou, Jimmy S Ren, Chunle Guo 和 Chongyi Li。Dit4sr：驯服扩散Transformer以实现真实世界图像超分辨率。载于*ICCV*，2025 年。[14]Ying Fan, Olivia Watkins, Yuqing Du, Hao Liu, Moonkyung Ryu, Craig Boutilier, Pieter Abbeel, Mohammad Ghavamzadeh, Kangwook Lee 和 Kimin Lee。Dpok：用于微调文本到图像扩散模型的强化学习。载于*NeurIPS*，2023 年。

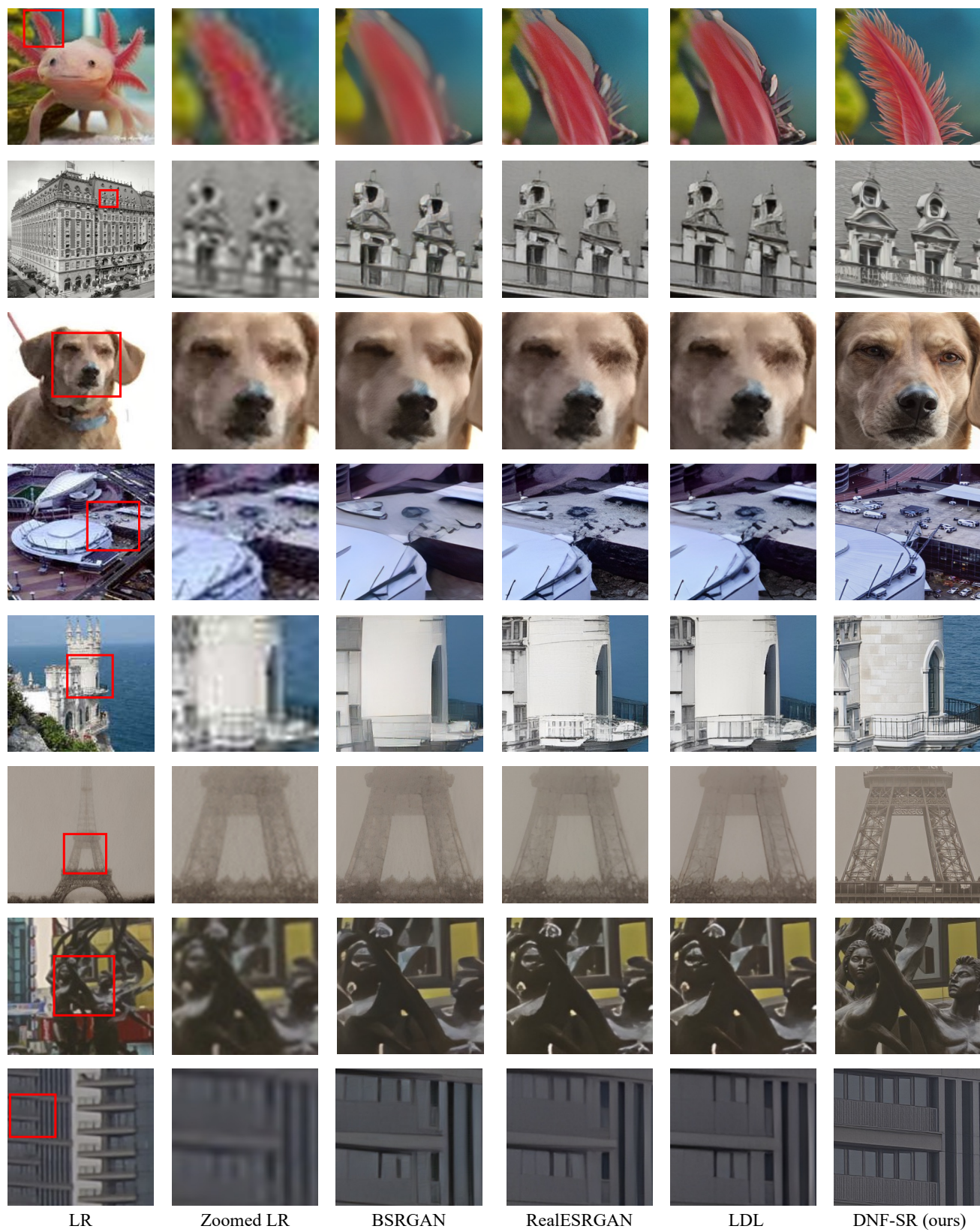


Figure 7. Vision comparisons between DNF-SR and GAN-based Real-ISR methods [23, 43, 61]. Zoom in for a better view.

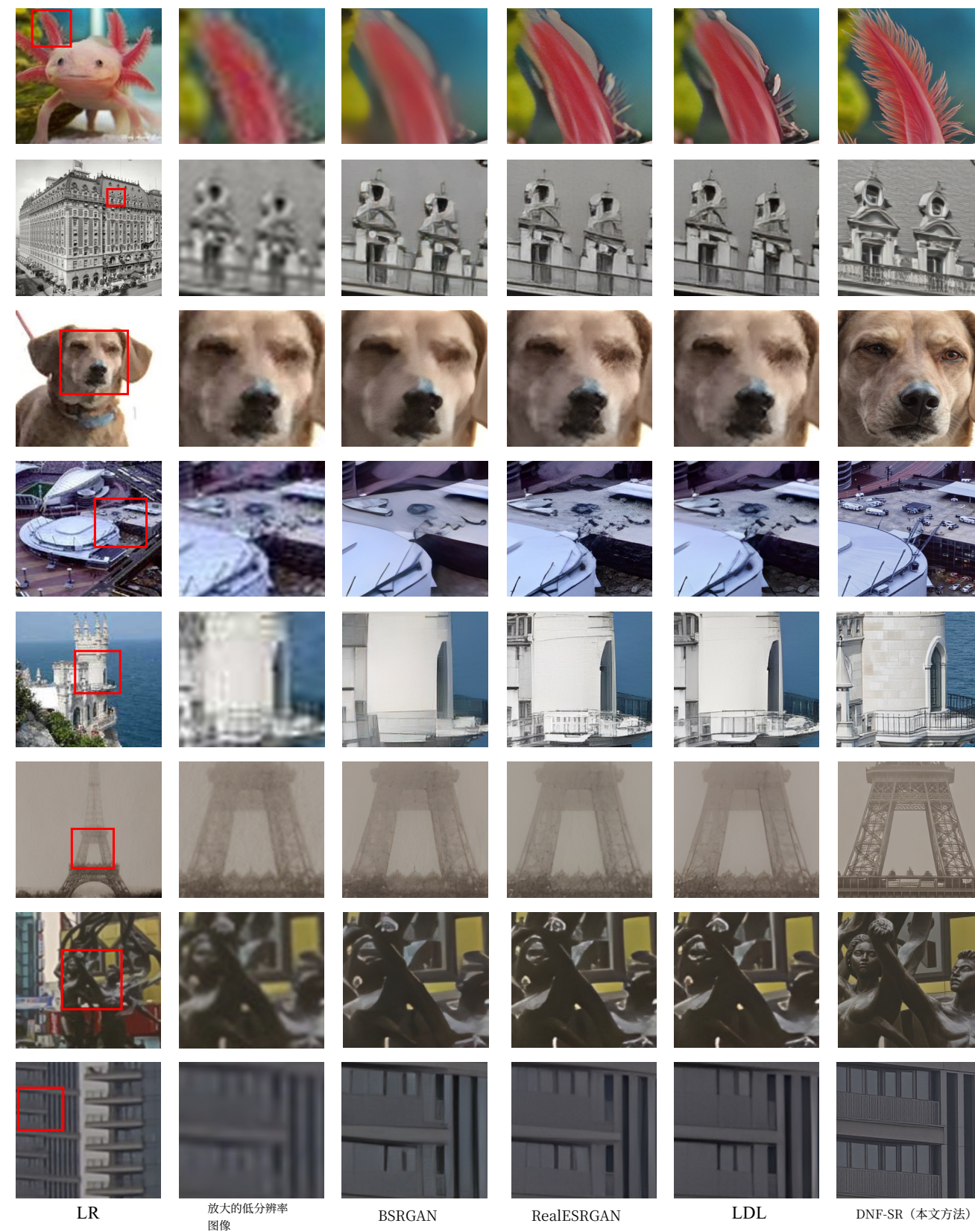
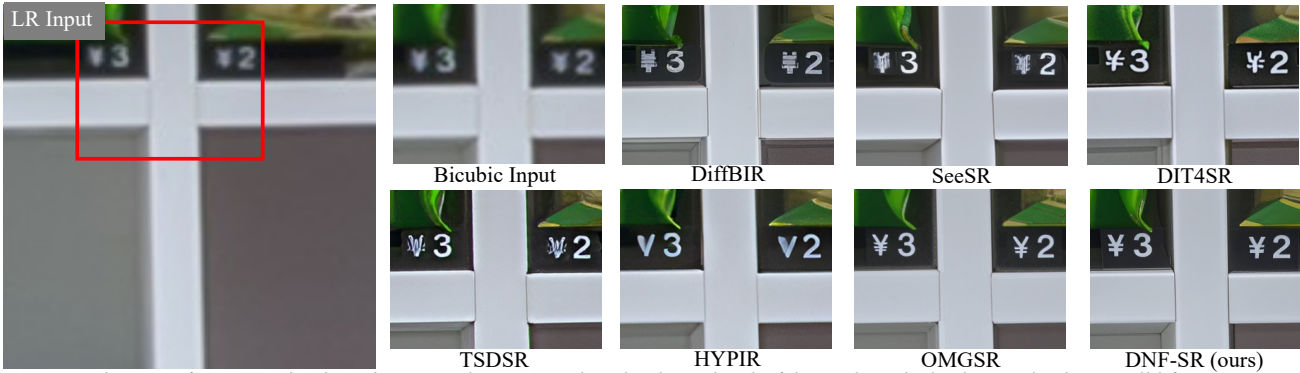
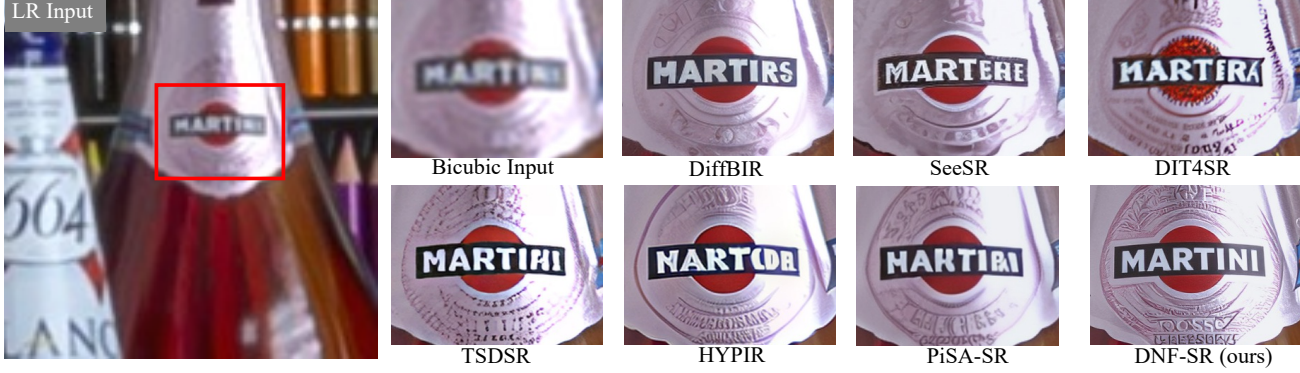


图7. DNF-SR与基于GAN的真实图像超分辨率方法的视觉效果对比 [23, 43, 61]。放大查看效果更佳。

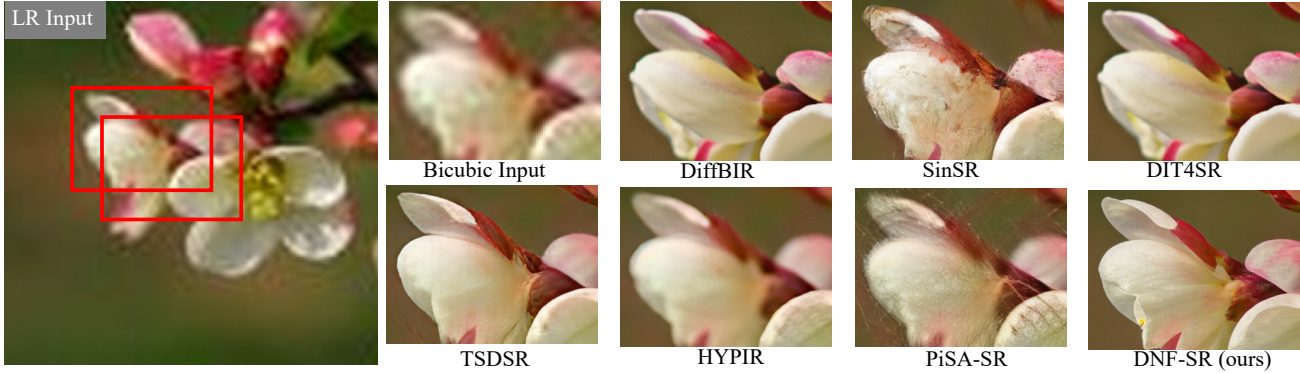
Caption: Close-up of a vending machine grid with compartments, some holding snacks priced ¥3 and ¥2, white dividers, and empty slots.



Caption: Close-up of a Martini bottle and a Kronenbourg 664 Blanc bottle, with colorful pencils in the background, vibrant still life.



Caption: Close-up of delicate white and pink flowering branches with a soft green background, featuring tender blossoms and buds in a serene.



Caption: Close-up of white chamomile flowers with yellow centers and green stems, against a softly blurred, warm-toned natural background.

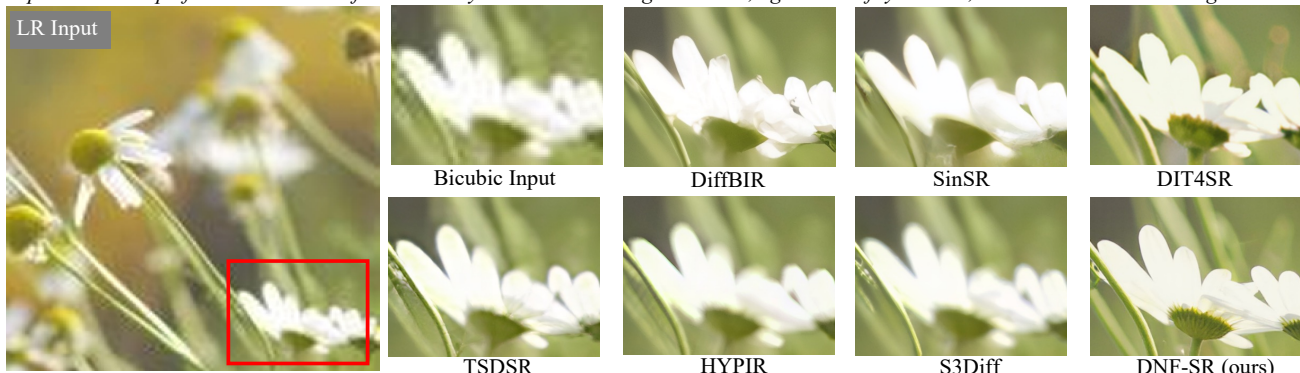
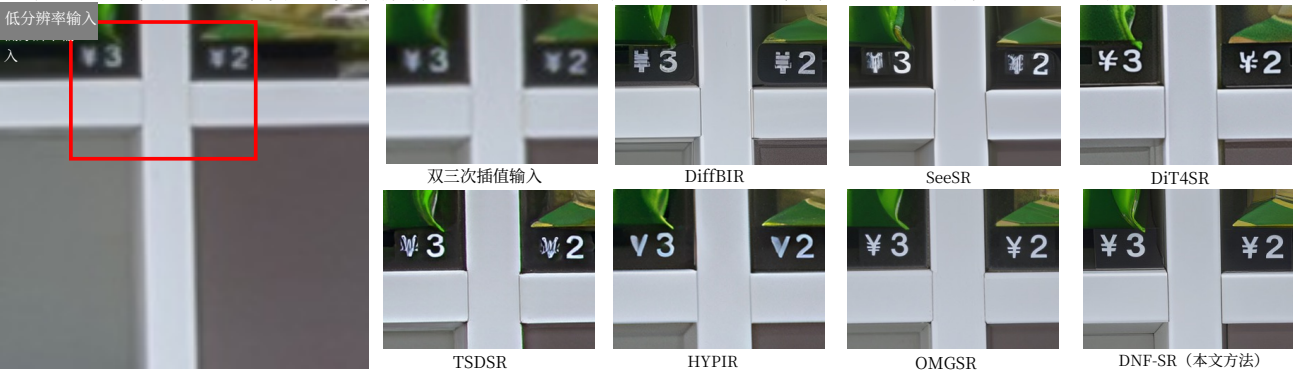


Figure 8. Vision comparisons between DNF-SR and different diffusion-based Real-ISR methods [12, 13, 24, 25, 38, 45, 50, 53, 60].

文本描述：一台自动售货机的货架特写，部分隔间放着标价3日元和2日元的零食，装有白色隔板，还有一些空位。



文本描述：马天尼酒瓶和凯旋1664白啤酒瓶的特写，背景是彩色铅笔，充满生气的静物。



文本描述：柔和的绿色背景衬托下，白色与粉色花枝的特写，花枝上带着娇嫩的花朵和花蕾，尽显宁静。

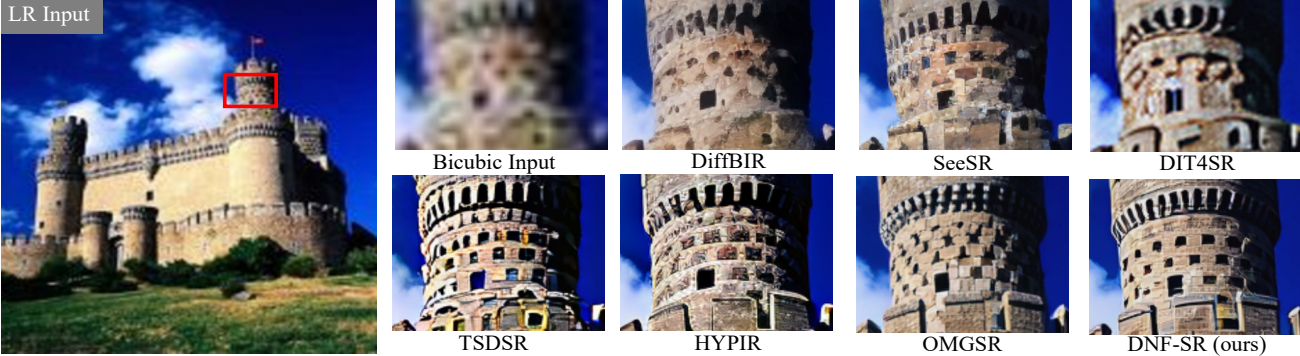


文本描述：白色洋甘菊花的特写，花心为黄色，茎为绿色，背景是柔焦的暖色调自然背景。

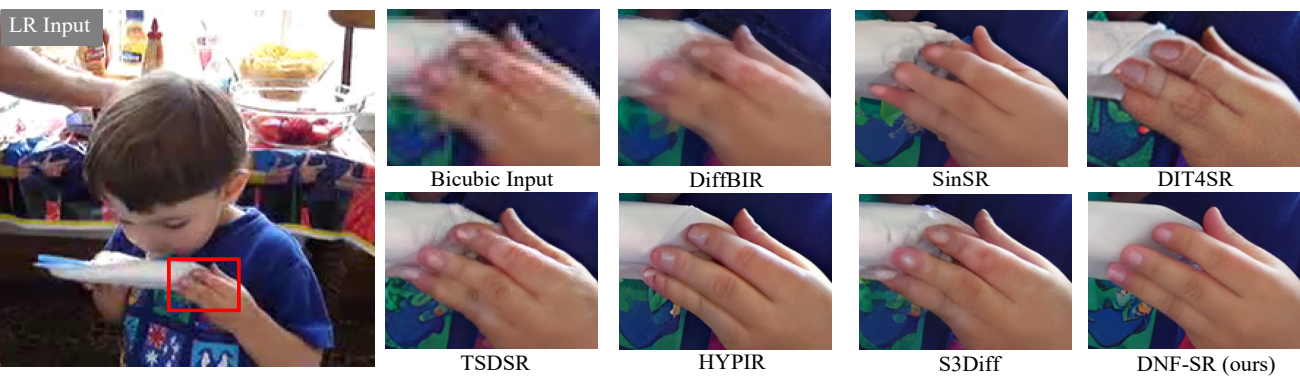


图8. DNF-SR与不同基于扩散的真实图像超分辨率方法的视觉效果对比 [12, 13, 24, 25, 38, 45, 50, 53, 60].

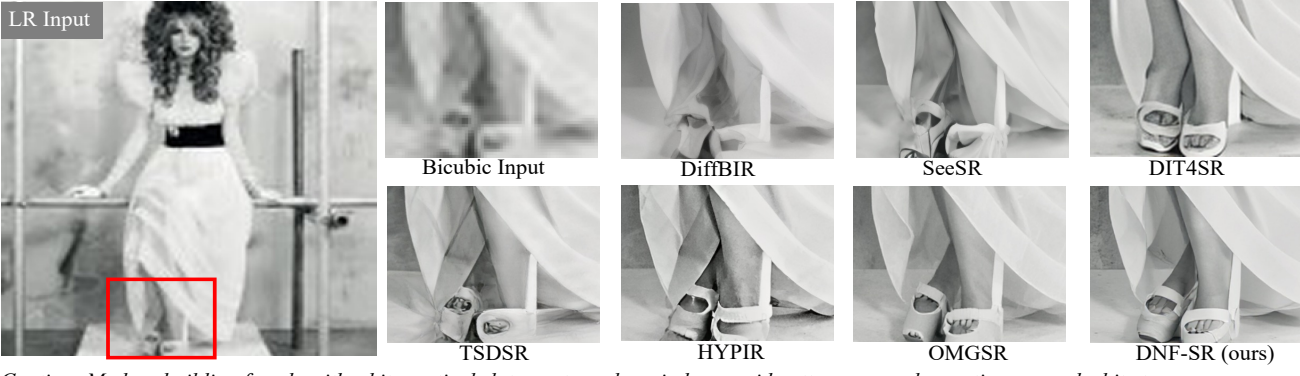
Caption: Medieval stone castle with tall towers, crenellated walls, flag on highest tower. Bright blue sky with scattered clouds, green grassy field, small bushes around.



Caption: Young child in blue shirt with colorful patterns, eating from white paper plate. Buffet table with red tomatoes, yellow chips, condiment bottles. Bright, casual family gathering scene.



Caption: Black and white vintage-style photo of a woman with voluminous curly hair, wearing a white dress with a black wide belt, leaning on metal rails.



Caption: Modern building facade with white vertical slats, rectangular windows, grid pattern, monochromatic gray and white tones.

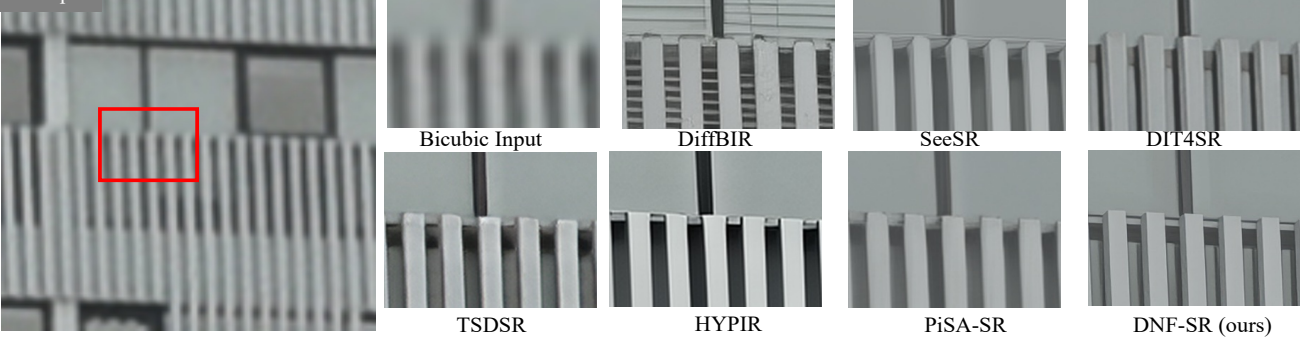
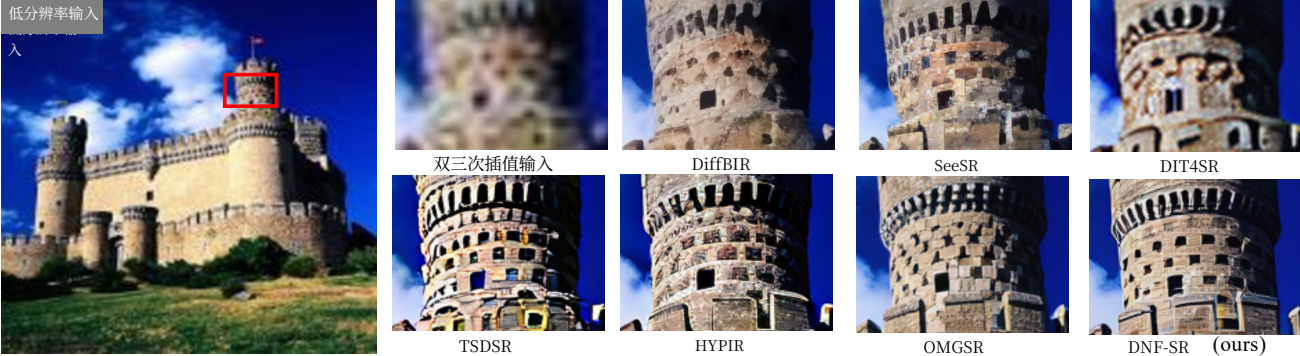
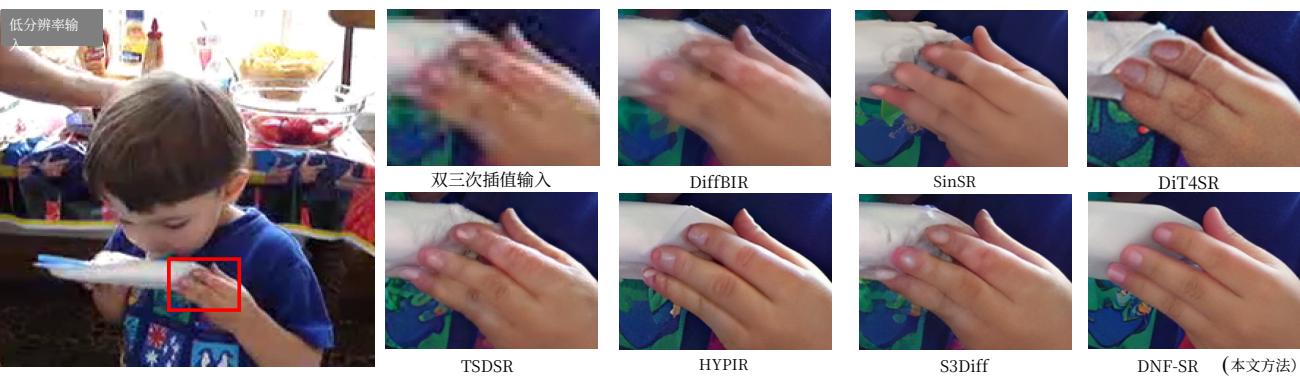


Figure 9. Vision comparisons between DNF-SR and different diffusion-based Real-ISR methods [12, 13, 24, 25, 38, 45, 50, 53, 60].

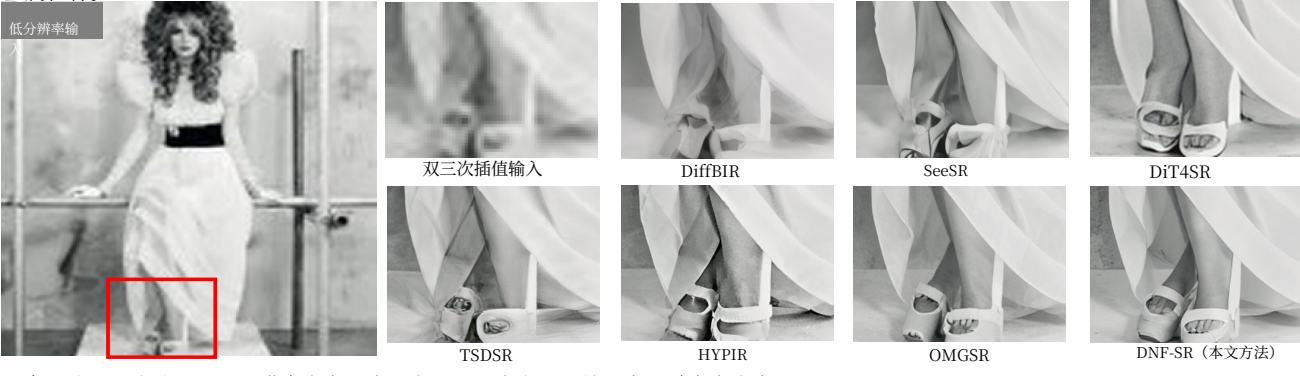
文本描述：中世纪石制城堡，有高耸的塔楼、雉堞墙，最高塔上悬挂旗帜。明亮的蓝天上飘着零散的云朵，绿色草地四周点缀着小灌木。



文本描述：幼童穿着带有彩色图案的蓝色衬衫，正从白色纸盘中取食。自助餐桌上有红番茄、黄色薯片和调味品瓶。明亮、轻松的家庭聚会场景。



文本描述：黑白复古风格的老照片，一位女子留着蓬松卷发，身穿白色连衣裙，系着黑色宽腰带，倚靠金属栏杆。



文本描述：现代建筑立面，带有白色竖直板条、矩形窗户、网格图案，单色灰白色调。

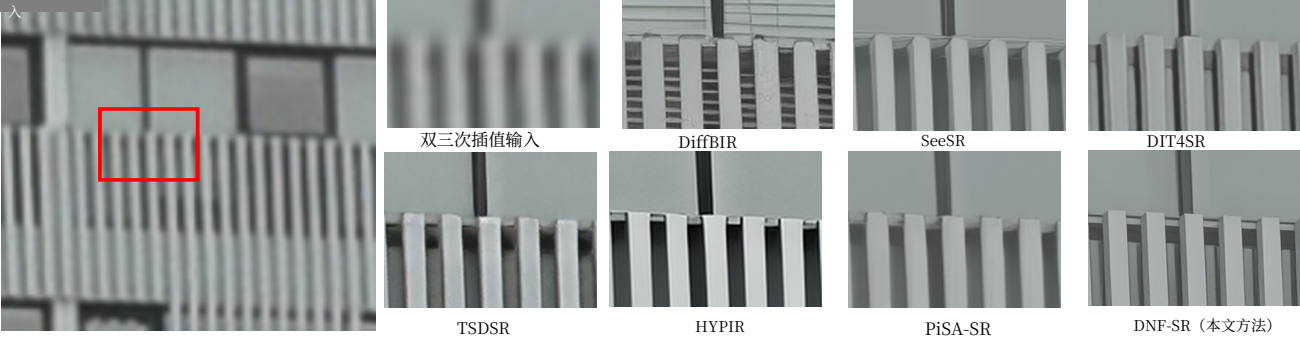


图 9. DNF-SR 与不同基于扩散的真实图像超分辨率方法的视觉效果对比 [12, 13, 24, 25, 38, 45, 50, 53, 60].

Algorithm 1: Training Procedure of Negativate-aware Feature Fine-tuning in DNF-SR		
Input: Training datasets $\{x_L, x_H, c\}$, fine-tuning one-step Diffusion-based SR Model including VAE encoder E_{ref} and velocity prediction network v_{ref} , pre-trained VAE decoder D_φ , number of samples K per training step, N raw reward functions $r^{raw}(\cdot) \in \mathbb{R}$, one fixed mid-timestep t_{mid} . Output: Post-trained one-step velocity prediction network v_θ for SR.		
1	Initialize data collection policy for velocity prediction $v_{old} \leftarrow v_{ref}$. Initialize training policy for velocity prediction $v_\theta \leftarrow v_{ref}$. Initialize data buffer $\mathcal{D} \leftarrow \emptyset$	while train do
	/* Rollout Step, Data Collection */	
2	for each sampled data $(x_L, x_H, c) \sim \mathcal{D}$ do	
3	Sample K standard Gaussian noises $\epsilon^{1:K}$ and collect K restored images $\hat{x}_H^{1:K}$ using v_{old} . Compute rewards $\{r_{1:N}^{raw}\}^{1:K}$ using N raw reward functions, respectively. Standardize raw rewards in group: $r_i^{std} := (r_i^{raw} - \text{mean}(\{r_i^{raw}\}^{1:K})) / \text{std}(\{r_i^{raw}\}^{1:K})$. Normalize rewards using the standard Gaussian cumulative distribution function: $r_i = \Phi(X < r^{std})$. Average the N rewards: $r = \text{avg}(r_{1:N})$ $D \leftarrow \{c, x_L, \hat{x}_H^{1:K}, r^{1:K}\}$	
4	end	
	/* Gradient Step, Policy Optimization */	
5	for each mini batch $\{c, x_L, \hat{x}_H, r\}$ do	
6	Encode the LR image: $z_L = E_{ref}(x_L)$. Forward diffusion process: $z_t = t_{mid}\epsilon + (1 - t_{mid})z_L$. /* Calculate Positive Optimization Direction */	
7	Implicit positive velocity: $v_\theta^+ := (1 - \beta)v^{old}(z_t, c, t_{mid}) + \beta v_\theta(z_t, c, t_{mid})$. Implicit positive image: $\hat{x}_\theta^+ := D_\varphi(z_t - t_{mid}v_\theta^+)$. Positive optimization direction: $\mathcal{L}_\theta^+ = r\mathcal{L}_{rec}(\hat{x}_\theta^+, \hat{x}_H)$. /* Calculate Positive Optimization Direction */	
8	Implicit negative velocity: $v_\theta^- := (1 + \beta)v^{old}(z_t, c, t_{mid}) - \beta v_\theta(z_t, c, t_{mid})$. Implicit negative image: $\hat{x}_\theta^- := D_\varphi(z_t - t_{mid}v_\theta^-)$. Negative optimization direction: $\mathcal{L}_\theta^- = (1 - r)\mathcal{L}_{rec}(\hat{x}_\theta^-, \hat{x}_H)$. /* Update Model Parameters */	
9	$\theta \leftarrow \theta - \lambda \nabla_\theta [\mathcal{L}_\theta^+ + \mathcal{L}_\theta^-]$	
10	end	
	/* Online Update */	
11	Update data collection policy $v_{old} \leftarrow v_\theta$, and clear buffer $\mathcal{D} \leftarrow \emptyset$.	
12	end	

[15] Ian J Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *NeurIPS*, 2014.

[16] Shuhang Gu, Andreas Lugmayr, Martin Danelljan, Manuel Fritsche, Julien Lamour, and Radu Timofte. Div8k: Diverse 8k resolution image dataset. In *ICCVW*, 2019.

[17] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *ArXiv preprint*, 2025.

[18] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. In *ICLR*, 2022.

[19] Gu Jinjin, Cai Haoming, Chen Haoyu, Ye Xiaoxing, Jimmy S Ren, and Dong Chao. Pipal: a large-scale image quality assessment dataset for perceptual image restoration. In *ECCV*, 2020.

[20] Tero Karras. A style-based generator architecture for gener-

ative adversarial networks. *ArXiv preprint*, 2019.

[21] Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. Musiq: Multi-scale image quality transformer. In *ICCV*, 2021.

[22] Jianze Li, Jiezhang Cao, Yong Guo, Wenbo Li, and Yulun Zhang. One diffusion step to real-world super-resolution via flow trajectory distillation. *ArXiv preprint*, 2025.

[23] Jie Liang, Hui Zeng, and Lei Zhang. Details or artifacts: A locally discriminative learning approach to realistic image super-resolution. In *CVPR*, 2022.

[24] Xinqi Lin, Jingwen He, Ziyang Chen, Zhaoyang Lyu, Bo Dai, Fanghua Yu, Yu Qiao, Wanli Ouyang, and Chao Dong. Diff-bir: Toward blind image restoration with generative diffusion prior. In *ECCV*, 2024.

[25] Xinqi Lin, Fanghua Yu, Jinfan Hu, Zhiyuan You, Wu Shi, Jimmy S Ren, Jinjin Gu, and Chao Dong. Harnessing diffusion-yielded score priors for image restoration. *ArXiv preprint*, 2025.

[26] Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *ICLR*, 2023.

算法1: DNF-SR中负激活感知特征微调的训练流程
输入 : 训练数据集 $\{x_L, x_H, c\}$, 待微调的单步基于扩散的超分辨率模型, 包括VAE编码器 E_{ref} 和速度预测网络 v_{ref} , 预训练VAE解码器 D_φ , 每个训练步的样本数 K , N 原始奖励函数 $r^{raw}(\cdot) \in \mathbb{R}$, 一个固定中间时间步 t_{mid} 。 输出 : 用于超分辨率的后训练单步速度预测网络 v_θ 。 1 初始化速度预测的数据收集策略 $v_{old} \leftarrow v_{ref}$ 。 初始化速度预测的训练策略 $v_\theta \leftarrow v_{ref}$ 。 初始化数据缓冲区 $\mathcal{D} \leftarrow \emptyset$ 。 当训练时执行 /* 展开步骤, 数据收集 */ 2 对于每个采样数据 $(x_L, x_H, c) \sim \mathcal{D}$ 执行 3 采样 K 标准高斯噪声 $\epsilon^{1:K}$, 并使用 v_{old} 收集 K 恢复图像 $\hat{x}_H^{1:K}$ 。 分别使用 N 原始奖励函数计算奖励 $\{r_{1:N}^{raw}\}^{1:K}$ 。 对组内原始奖励进行标准化: $r_i^{std} = (r_i^{raw} - \text{mean}(\{r_i^{raw}\}^{1:K})) / \text{std}(\{r_i^{raw}\}^{1:K})$ 。 使用标准高斯累积分布函数归一化奖励: $r_i = \Phi(X < r^{std})$ 。 对 N 个奖励取平均: $r = \text{avg}(r_{1:N})$ 。 $D \leftarrow \{c, x_L, \hat{x}_H^{1:K}, r^{1:K}\}$ 4 结束 /* 梯度步骤, 策略优化 */ 5 对于每个小批量 $\{c, x_L, \hat{x}_H, r\}$ 执行 6 编码低分辨率图像: $z_L = E_{ref}(x_L)$ 。 前向扩散过程: $z_t = t_{mid}\epsilon + (1 - t_{mid})z_L$ 。 /* 计算正优化方向 */ 7 隐式正速度: $v_\theta^+ = (1 - \beta)v^{old}(z_t, c, t_{mid}) + \beta v_\theta(z_t, c, t_{mid})$ 。 隐式正图像: $\hat{x}_\theta^+ = D_\varphi(z_t - t_{mid}v_\theta^+)$ 。 正优化方向: $\mathcal{L}_\theta^+ = r\mathcal{L}_{rec}(\hat{x}_\theta^+, \hat{x}_H)$ 。 /* 计算负优化方向 */ 8 隐式负速度: $v_\theta^- = (1 + \beta)v^{old}(z_t, c, t_{mid}) - \beta v_\theta(z_t, c, t_{mid})$ 。 隐式负图像: $\hat{x}_\theta^- = D_\varphi(z_t - t_{mid}v_\theta^-)$ 。 负优化方向: $\mathcal{L}_\theta^- = (1 - r)\mathcal{L}_{rec}(\hat{x}_\theta^-, \hat{x}_H)$ 。 /* 更新模型参数 */ $\theta \leftarrow \theta - \lambda \nabla_\theta [\mathcal{L}_\theta^+ + \mathcal{L}_\theta^-]$ 10 结束 /* 在线更新 */ 11 更新数据收集策略 $v_{old} \leftarrow v_\theta$, 并清空缓冲区 $\mathcal{D} \leftarrow \emptyset$ 。 12 结束

[15] Ian J Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 生成对抗网络。In *NeurIPS*, 2014.[16] Shuhang Gu, Andreas Lugmayr, Martin Danelljan, Manuel Fritsche, Julien Lamour, and Radu Timofte. Div8k: 多样化8k分辨率图像数据集。In *ICCVW*, 2019.[17] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: 通过强化学习激励大语言模型的推理能力。 *arXiv* 预印本, 2025.[18] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. LoRA: 大语言模型的低秩适应。 In *ICLR*, 2022.[19] Gu Jinjin, Cai Haoming, Chen Haoyu, Ye Xiaoxing, Jimmy S Ren, and Dong Chao. Pipal: 用于感知图像复原的大规模图像质量评估数据集。 In *ECCV*, 2020.[20] Tero Karras. 一种基于风格的生成器架构用于生成-

对抗生成网络。 *arXiv* 预印本, 2019年。[21] Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. Musiq: 多尺度图像质量Transformer。 见 *ICCV*, 2021年。[22] Jianze Li, Jiezhang Cao, Yong Guo, Wenbo Li, and Yulun Zhang. 通过流轨迹蒸馏实现一步扩散的真实世界超分辨率。 *arXiv* 预印本, 2025年。[23] Jie Liang, Hui Zeng, and Lei Zhang. 细节还是伪影: 一种面向真实图像超分辨率的局部判别学习方法。 见 *CVPR*, 2022年。[24] Xinqi Lin, Jingwen He, Ziyang Chen, Zhaoyang Lyu, Bo Dai, Fanghua Yu, Yu Qiao, Wanli Ouyang, and Chao Dong. Diff-bir: 利用生成扩散先验实现盲图像复原。 见 *ECCV*, 2024年。[25] Xinqi Lin, Fanghua Yu, Jinfan Hu, Zhiyuan You, Wu Shi, Jimmy S Ren, Jinjin Gu, and Chao Dong. 利用扩散产生的分数先验进行图像复原。 *arXiv* 预印本, 2025年。[26] Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. 用于生成建模的流匹配。 见 *ICLR*, 2023年。

- [27] Jie Liu, Gongye Liu, Jiajun Liang, Yangguang Li, Jiaheng Liu, Xintao Wang, Pengfei Wan, Di Zhang, and Wanli Ouyang. Flow-grpo: Training flow matching models via on-line rl. *ArXiv preprint*, 2025.
- [28] Xingchao Liu, Chengyue Gong, et al. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *ICLR*, 2023.
- [29] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *ArXiv preprint*, 2017.
- [30] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *NeurIPS*, 2022.
- [31] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *ICCV*, 2023.
- [32] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. Sdxl: Improving latent diffusion models for high-resolution image synthesis. *ArXiv preprint*, 2023.
- [33] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *NeurIPS*, 2023.
- [34] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, 2022.
- [35] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily L Denton, Kamyar Ghasemipour, Raphael Gontijo Lopes, Burcu Karagol Ayan, Tim Salimans, et al. Photorealistic text-to-image diffusion models with deep language understanding. In *NeurIPS*, 2022.
- [36] Chao Song, Shaobang Li, Frederick WB Li, and Bailin Yang. Wdfr: Normalizing flow based on the wavelet-domain for super-resolution. *Computational Visual Media*, 2025.
- [37] Lingchen Sun, Rongyuan Wu, Jie Liang, Zhengqiang Zhang, Hongwei Yong, and Lei Zhang. Improving the stability and efficiency of diffusion models for content consistent super-resolution. *ArXiv preprint*, 2023.
- [38] Lingchen Sun, Rongyuan Wu, Zhiyuan Ma, Shuaizheng Liu, Qiaosi Yi, and Lei Zhang. Pixel-level and semantic-level adjustable super-resolution: A dual-lora approach. In *CVPR*, 2025.
- [39] Lingchen Sun, Rongyuan Wu, Zhiyuan Ma, Shuaizheng Liu, Qiaosi Yi, and Lei Zhang. Pixel-level and semantic-level adjustable super-resolution: A dual-lora approach. In *CVPR*, 2025.
- [40] Radu Timofte, Eirikur Agustsson, Luc Van Gool, Ming-Hsuan Yang, and Lei Zhang. Ntire 2017 challenge on single image super-resolution: Methods and results. In *CVPRW*, 2017.
- [41] Bram Wallace, Meihua Dang, Rafael Rafailov, Linqi Zhou, Aaron Lou, Senthil Purushwalkam, Stefano Ermon, Caiming Xiong, Shafiq Joty, and Nikhil Naik. Diffusion model alignment using direct preference optimization. In *CVPR*, 2024.
- [42] Jianyi Wang, Kelvin CK Chan, and Chen Change Loy. Exploring clip for assessing the look and feel of images. In *AAAI*, 2023.
- [43] Xintao Wang, Liangbin Xie, Chao Dong, and Ying Shan. Real-esrgan: Training real-world blind super-resolution with pure synthetic data. In *ICCV*, 2021.
- [44] Xin Wang, Jing-Ke Yan, Jing-Ye Cai, Jian-Hua Deng, Qin Qin, and Yao Cheng. Super-resolution reconstruction of single image for latent features. *Computational Visual Media*, 2024.
- [45] Yufei Wang, Wenhan Yang, Xinyuan Chen, Yaohui Wang, Lanqing Guo, Lap-Pui Chau, Ziwei Liu, Yu Qiao, Alex C Kot, and Bihan Wen. Sinsr: diffusion-based image super-resolution in a single step. In *CVPR*, 2024.
- [46] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: from error visibility to structural similarity. *TIP*, 2004.
- [47] Pengxu Wei, Ziwei Xie, Hannan Lu, Zongyuan Zhan, Qixiang Ye, Wangmeng Zuo, and Liang Lin. Component divide-and-conquer for real-world image super-resolution. In *ECCV*, 2020.
- [48] Haoning Wu, Zicheng Zhang, Weixia Zhang, Chaofeng Chen, Chunyi Li, Liang Liao, Annan Wang, Erli Zhang, Wenxiu Sun, Qiong Yan, Xiongkuo Min, Guangtai Zhai, and Weisi Lin. Q-align: Teaching lmms for visual scoring via discrete text-defined levels. In *ICML*, 2024.
- [49] Rongyuan Wu, Lingchen Sun, Zhiyuan Ma, and Lei Zhang. One-step effective diffusion network for real-world image super-resolution. *ArXiv preprint*, 2024.
- [50] Rongyuan Wu, Tao Yang, Lingchen Sun, Zhengqiang Zhang, Shuai Li, and Lei Zhang. Seesr: Towards semantics-aware real-world image super-resolution. In *CVPR*, 2024.
- [51] Shixiang Wu, Chao Dong, and Yu Qiao. Exploring contextual priors for real-world image super-resolution. *Computational Visual Media*, 2025.
- [52] Tianhe Wu, Jian Zou, Jie Liang, Lei Zhang, and Kede Ma. Visualquality-rl: Reasoning-induced image quality assessment via reinforcement learning to rank. *ArXiv preprint*, 2025.
- [53] Zhiqiang Wu, Zhaomang Sun, Tong Zhou, Bingtao Fu, Ji Cong, Yitong Dong, Huaqi Zhang, Xuan Tang, Mingsong Chen, and Xian Wei. Omgsr: You only need one mid-timestep guidance for real-world image super-resolution. *ArXiv preprint*, 2025.
- [54] Rui Xie, Chen Zhao, Kai Zhang, Zhenyu Zhang, Jun Zhou, Jian Yang, and Ying Tai. Addsr: Accelerating diffusion-based blind super-resolution with adversarial diffusion distillation. *ArXiv preprint*, 2024.
- [55] Zeyue Xue, Jie Wu, Yu Gao, Fangyuan Kong, Lingting Zhu, Mengzhao Chen, Zhiheng Liu, Wei Liu, Qiushan Guo, Weilin Huang, et al. Dancegrpo: Unleashing grpo on visual generation. *ArXiv preprint*, 2025.
- [56] Jian Yang, Jiayao Xu, Chi Do-Kim Pham, and Jinjia Zhou. Jvcsr+: Adaptively learned video compressive sensing reconstruction with joint in-loop reference enhancement and out-loop super-resolution. *Computational Visual Media*, 2025.
- [57] Sidi Yang, Tianhe Wu, Shuwei Shi, Shanshan Lao, Yuan Gong, Mingdeng Cao, Jiahao Wang, and Yujiu Yang. Maniq: Multi-dimension attention network for no-reference image quality assessment. In *CVPR*, 2022.
- [27] Jie Liu, Gongye Liu, Jiajun Liang, Yangguang Li, Jiaheng Liu, Xintao Wang, Pengfei Wan, Di Zhang, and Wanli Ouyang. Flow-grpo: 通过在线强化学习训练流匹配模型. *arXiv* 预印本, 2025年.
- [28] Xingchao Liu, Chengyue Gong, et al. Flow straight and fast: 利用矫正流学习生成和迁移数据. 见*ICLR*, 2023年.
- [29] Ilya Loshchilov and Frank Hutter. 解耦权重衰减正则化. *arXiv* 预印本, 2017年.
- [30] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 训练语言模型遵循人类反馈的指令. 见*NeurIPS*, 2022年.
- [31] William Peebles and Saining Xie. 基于Transformer的可扩展扩散模型. 见*ICCV*, 2023年.
- [32] Dustin Podell, Zio n English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas M üller, Joe Penna, and Robin Rombach. SDXL: 改进潜空间扩散模型以实现高分辨率图像合成. *arXiv* 预印本, 2023年.
- [33] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 直接偏好优化: 你的语言模型其实是一个奖励模型. 见*NeurIPS*, 2023年.
- [34] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Bj örn Ommer. 利用潜空间扩散模型实现高分辨率图像合成. 见*CVPR*, 2022年.
- [35] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily L Denton, Kamyar Ghasemipour, Raphael Gontijo Lopes, Burcu Karagol Ayan, Tim Salimans, et al. 具有深度语言理解的逼真文本到图像扩散模型. 见*NeurIPS*, 2022年.
- [36] Chao Song, Shaobang Li, Frederick WB Li, and Bailin Yang. Wdfr: 基于小波域的归一化流用于超分辨率. 计算可视媒体, 2025年.
- [37] Lingchen Sun, Rongyuan Wu, Jie Liang, Zhengqiang Zhang, Hongwei Yong, and Lei Zhang. 提高内容一致超分辨率扩散模型的稳定性和效率. *arXiv* 预印本, 2023年.
- [38] Lingchen Sun, Rongyuan Wu, Zhiyuan Ma, Shuaizheng Liu, Qiaosi Yi, and Lei Zhang. 像素级和语义级可调超分辨率: 一种双LoRA方法. 见*CVPR*, 2025年.
- [39] Lingchen Sun, Rongyuan Wu, Zhiyuan Ma, Shuaizheng Liu, Qiaosi Yi, and Lei Zhang. 像素级和语义级可调超分辨率: 一种双LoRA方法. 见*CVPR*, 2025年.
- [40] Radu Timofte, Eirikur Agustsson, Luc Van Gool, Ming-Hsuan Yang, and Lei Zhang. Ntire 2017 挑战赛单图像超分辨率: 方法及结果. 见*CVPRW*, 2017年.
- [41] Bram Wallace, Meihua Dang, Rafael Rafailov, Linqi Zhou, Aaron Lou, Senthil Purushwalkam, Stefano Ermon, Caiming Xiong, Shafiq Joty, and Nikhil Naik. 使用直接偏好优化进行扩散模型对齐. 见*CVPR*, 2024年.
- [42] Jianyi Wang, Kelvin CK Chan, and Chen Change Loy. 探索 CLIP用于评估图像观感. 见*AAAI*, 2023年.
- [43] Xintao Wang, Liangbin Xie, Chao Dong, and Ying Shan. Real-esrgan: 使用纯合成数据训练真实世界盲超分辨率. In *ICCV*, 2021.[44] Xin Wang, Jing-Ke Yan, Jing-Ye Cai, Jian-Hua Deng, Qin Qin, and Yao Cheng. 单幅图像潜变量特征超分辨率重建. 计算可视媒体, 2024.[45] Yufei Wang, Wenhan Yang, Xinyuan Chen, Yaohui Wang, Lanqing Guo, Lap-Pui Chau, Ziwei Liu, Yu Qiao, Alex C Kot, and Bihan Wen. Sinsr: 单步基于扩散的图像超分辨率. In *CVPR*, 2024.[46] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. 图像质量评估: 从误差可见性到结构相似性. *TIP*, 2004.[47] Pengxu Wei, Ziwei Xie, Hannan Lu, Zongyuan Zhan, Qixiang Ye, Wangmeng Zuo, and Liang Lin. 面向真实世界图像超分辨率的组件分而治之. In *ECCV*, 2020.[48] Haoning Wu, Zicheng Zhang, Weixia Zhang, Chaofeng Chen, Chunyi Li, Liang Liao, Annan Wang, Erli Zhang, Wenxiu Sun, Qiong Yan, Xiongkuo Min, Guangtai Zhai, and Weisi Lin. Q-align: 通过离散文本定义级别教授大语言多模态模型进行视觉评分. In *ICML*, 2024.[49] Rongyuan Wu, Lingchen Sun, Zhiyuan Ma, and Lei Zhang. 面向真实世界图像超分辨率的单步有效扩散网络. *arXiv* 预印本, 2024.[50] Rongyuan Wu, Tao Yang, Lingchen Sun, Zhengqiang Zhang, Shuai Li, and Lei Zhang. Seesr: 面向语义感知的真实世界图像超分辨率. In *CVPR*, 2024.[51] Shixiang Wu, Chao Dong, and Yu Qiao. 探索真实世界图像超分辨率的上下文先验. 计算可视媒体, 2025.[52] Tianhe Wu, Jian Zou, Jie Liang, Lei Zhang, and Kede Ma. Visualquality-rl: 通过强化学习排序实现推理驱动的图像质量评估. *arXiv* 预印本, 2025.[53] Zhiqiang Wu, Zhaomang Sun, Tong Zhou, Bingtao Fu, Ji Cong, Yitong Dong, Huaqi Zhang, Xuan Tang, Mingsong Chen, and Xian Wei. Omgsr: 真实世界图像超分辨率仅需一个中间时间步引导. *arXiv* 预印本, 2025.[54] Rui Xie, Chen Zhao, Kai Zhang, Zhenyu Zhang, Jun Zhou, Jian Yang, and Ying Tai. Addsr: 利用对抗扩散蒸馏加速基于扩散的盲超分辨率. *arXiv* 预印本, 2024.[55] Zeyue Xue, Jie Wu, Yu Gao, Fangyuan Kong, Lingting Zhu, Mengzhao Chen, Zhiheng Liu, Wei Liu, Qiushan Guo, Weilin Huang, et al. Dancegrpo: 在视觉生成中释放 GRPO. *arXiv* 预印本, 2025.[56] Jian Yang, Jiayao Xu, Chi Do-Kim Pham, and Jinjia Zhou. Jvcsr+: 联合环内参考增强与环外超分辨率的自适应视频压缩感知重建. 计算可视媒体, 2025.[57] Sidi Yang, Tianhe Wu, Shuwei Shi, Shanshan Lao, Yuan Gong, Mingdeng Cao, Jiahao Wang, and Yujiu Yang. Maniq: 用于无参考图像质量评估的多维注意力网络. In *CVPR*, 2022.

[58] Fanghua Yu, Jinjin Gu, Zheyuan Li, Jinfan Hu, Xiangtao Kong, Xintao Wang, Jingwen He, Yu Qiao, and Chao Dong. Scaling up to excellence: Practicing model scaling for photo-realistic image restoration in the wild. In *CVPR*, 2024.

[59] Zongsheng Yue, Jianyi Wang, and Chen Change Loy. Resshift: Efficient diffusion model for image super-resolution by residual shifting. In *NeurIPS*, 2024.

[60] Aiping Zhang, Zongsheng Yue, Renjing Pei, Wenqi Ren, and Xiaochun Cao. Degradation-guided one-step image super-resolution with diffusion priors. *ArXiv preprint*, 2024.

[61] Kai Zhang, Jingyun Liang, Luc Van Gool, and Radu Timofte. Designing a practical degradation model for deep blind image super-resolution. In *ICCV*, 2021.

[62] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *CVPR*, 2023.

[63] Tainyi Zhang, Zheng-Peng Duan, Peng-Tao Jiang, Bo Li, Ming-Ming Cheng, Chun-Le Guo, and Chongyi Li. Time-aware one step diffusion network for real-world image super-resolution. *ArXiv preprint*, 2025.

[64] Kaiwen Zheng, Huayu Chen, Haotian Ye, Haoxiang Wang, Qinsheng Zhang, Kai Jiang, Hang Su, Stefano Ermon, Jun Zhu, and Ming-Yu Liu. Diffusionnft: Online diffusion reinforcement with forward process. *ArXiv preprint*, 2025.

[58] Fanghua Yu, Jinjin Gu, Zheyuan Li, Jinfan Hu, Xiangtao Kong, Xintao Wang, Jingwen He, Yu Qiao, and Chao Dong. 扩展至卓越：在真实场景中实践模型缩放以实现照片级图像恢复. In *CVPR*, 2024.[59]Zongsheng Yue, Jianyi Wang, and Chen Change Loy. Resshift: 通过残差移位实现图像超分辨率的高效扩散模型. In *NeurIPS*, 2024.[60] Aiping Zhang, Zongsheng Yue, Renjing Pei, Wenqi Ren, and Xiaochun Cao. 利用扩散先验的退化引导单步图像超分辨率. *arXiv* 预印本, 2024.[61] Kai Zhang, Jingyun Liang, Luc Van Gool, and Radu Timofte. 为深度盲图像超分辨率设计实用退化模型. In *ICCV*, 2021.[62] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. 为文生图扩散模型添加条件控制. In *CVPR*, 2023.[63] Tainyi Zhang, Zheng-Peng Duan, Peng-Tao Jiang, Bo Li, Ming-Ming Cheng, Chun-Le Guo, and Chongyi Li. 面向真实世界图像超分辨率的时间感知单步扩散网络. *arXiv* 预印本, 2025.[64] Kaiwen Zheng, Huayu Chen, Haotian Ye, Haoxiang Wang, Qinsheng Zhang, Kai Jiang, Hang Su, Stefano Ermon, Jun Zhu, and Ming-Yu Liu. Diffusionnft: 基于前向过程的在线扩散强化. *arXiv* 预印本, 2025.